

SGA 3, Exposé XX
Reductive Groups of Semisimple Rank 1
English Reconstruction

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1 Elementary systems. The groups U_α and $U_{-\alpha}$

Recollection 1.1. *Let $S = \operatorname{Spec}(k)$, where k is an algebraically closed field, and let G be a reductive S -group of semisimple rank one and T a maximal torus of G , necessarily split. Then*

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

where α and $-\alpha$ are the roots of G with respect to T . Moreover, there exist two group monomorphisms

$$p_\alpha: \mathbb{G}_{a,S} \longrightarrow G, \quad p_{-\alpha}: \mathbb{G}_{a,S} \longrightarrow G$$

such that

$$t p_\alpha(x) t^{-1} = p_\alpha(\alpha(t)x), \quad t p_{-\alpha}(x) t^{-1} = p_{-\alpha}(\alpha(t)^{-1}x)$$

for every $S' \rightarrow S$, $t \in T(S')$, and $x \in \mathbb{G}_a(S')$, and such that the morphism

$$\mathbb{G}_{a,S} \times_S T \times_S \mathbb{G}_{a,S} \longrightarrow G, \quad (y, t, x) \longmapsto p_{-\alpha}(y) t p_\alpha(x),$$

is radicial and dominant (Bible, Section 13.4, Corollary 2 to Theorem 3).

Since its tangent map at the identity is bijective, this morphism is also separable, hence birational. Zariski's Main Theorem (EGA III₁, 4.4.9) therefore shows that it is an open immersion.

Lemma 1.2. *Let S be a scheme, G an S -group scheme, T a torus of G , Q a subtorus of T , and α a character of T whose restriction to Q is nontrivial on every fiber. Let*

$$p_\alpha: \mathbb{G}_{a,S} \longrightarrow G \quad \text{and} \quad p_{-\alpha}: \mathbb{G}_{a,S} \longrightarrow G$$

be group morphisms normalized by T , with multipliers α and $-\alpha$, respectively. Suppose that the morphism

$$u: \mathbb{G}_{a,S} \times_S T \times_S \mathbb{G}_{a,S} \longrightarrow G, \quad u(y, t, x) = p_{-\alpha}(y) t p_\alpha(x),$$

defined on points by the displayed formula, is an open immersion. Finally, let $q \geq 0$ be an integer and let

$$p: \mathbb{G}_{a,S} \longrightarrow G$$

be a group morphism such that, for every $S' \rightarrow S$, $t \in Q(S')$, and $x \in \mathbb{G}_a(S')$,

$$\operatorname{int}(t)^q(p(x)) = p(\alpha(t)x).$$

Then there is a unique $\nu \in \mathbb{G}_a(S)$ such that

$$p(x) = p_\alpha(\nu x^q).$$

Indeed, let Ω be the image of u , and put $U = p^{-1}(\Omega)$. This is an open subset of $\mathbb{G}_{a,S}$ containing the zero section. For every section t of Q , multiplication by $\alpha(t)$ leaves U globally invariant. We have $U = \mathbb{G}_{a,S}$. It is enough to check this when S is the spectrum of an algebraically closed field k ; then

$$\alpha: Q(k) \longrightarrow k^\times$$

is surjective, so $U(k) \supset k^\times$, and hence $U = \mathbb{G}_{a,k}$.

⁰Editors' note: version of 13 October 2024.

There are consequently three morphisms

$$a: \mathbb{G}_{a,S} \longrightarrow \mathbb{G}_{a,S}, \quad b: \mathbb{G}_{a,S} \longrightarrow T, \quad c: \mathbb{G}_{a,S} \longrightarrow \mathbb{G}_{a,S}$$

such that

$$p(x) = p_{-\alpha}(a(x)) b(x) p_{\alpha}(c(x)).$$

The condition on p becomes

$$a(\alpha(t)x) = \alpha(t)^{-q}a(x), \quad b(\alpha(t)x) = b(x), \quad c(\alpha(t)x) = \alpha(t)^q c(x).$$

For the same reason, for every $S' \rightarrow S$ and $z \in \mathbb{G}_m(S')$,

$$a(zx) = z^{-q}a(x), \quad b(zx) = b(x), \quad c(zx) = z^q c(x),$$

and therefore

$$z^q a(z) = a(1), \quad b(z) = b(1), \quad c(z) = z^q c(1).$$

Since $\mathbb{G}_{m,S}$ is schematically dense in $\mathbb{G}_{a,S}$, for every $S' \rightarrow S$ and $x \in \mathbb{G}_a(S')$ we obtain

$$\begin{aligned} x^q a(x) &= a(1) = a(0) = 0, & \text{hence } a &= 0, \\ c(x) &= x^q c(1) = \nu x^q, & \text{for some } \nu &\in \mathbb{G}_a(S), \\ b(x) &= b(1) = b(0) = e, & \text{hence } b &= e. \end{aligned}$$

This completes the proof.

Definition 1.3. *An S -elementary system is a triple (G, T, α) in which*

- (i) G is a reductive S -group of semisimple rank one (Exposé XIX, 2.7);
- (ii) T is a maximal torus of G ;
- (iii) α is a root of G with respect to T (Exposé XIX, 3.2).

Thus there is a direct-sum decomposition (Exposé XIX, 3.5)

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^{\alpha} \oplus \mathfrak{g}^{-\alpha},$$

where $\mathfrak{g}^{\alpha}$ and $\mathfrak{g}^{-\alpha}$ are locally free of rank one.¹

1.4. If (G, T, α) is an S -elementary system, then $(G_{S'}, T_{S'}, \alpha_{S'})$ is an S' -elementary system for every $S' \rightarrow S$. If (G, T, α) is an S -elementary system, then so is $(G, T, -\alpha)$.

If S is a scheme, G a reductive S -group, T a maximal torus of G , and α a root of G with respect to T , then (Z_{α}, T, α) is an S -elementary system (Exposé XIX, 3.9).

Let (G, T, α) be an S -elementary system. The invertible module $\mathfrak{g}^{\alpha}$ is canonically endowed with a T -module structure. Hence the vector bundle $W(\mathfrak{g}^{\alpha})$ is also a T -module. On the other hand, the inner automorphisms by elements of T make G a group equipped with a group of operators T .

¹As $\mathcal{O}_S$ -modules.

Theorem 1.5. *Let (G, T, α) be an S -elementary system.*

- (i) *There exists a unique morphism of groups with operator group T*

$$\exp: W(\mathfrak{g}^\alpha) \longrightarrow G$$

whose induced morphism on Lie algebras is the canonical map $\mathfrak{g}^\alpha \rightarrow \mathfrak{g}$.² Equivalently, $\exp$ is the unique morphism such that, for every $S' \rightarrow S$, $t \in T(S')$, and $X, X' \in W(\mathfrak{g}^\alpha)(S')$,

$$\begin{aligned} \exp(X + X') &= \exp(X) \exp(X'), \\ \text{int}(t)(\exp(X)) &= \exp(\alpha(t)X), \\ \text{Lie}(\exp)(X) &= X. \end{aligned}$$

- (ii) *Define analogously, in the S -elementary system $(G, T, -\alpha)$,*

$$\exp: W(\mathfrak{g}^{-\alpha}) \longrightarrow G.$$

Then the morphism

$$W(\mathfrak{g}^{-\alpha}) \times_S T \times_S W(\mathfrak{g}^\alpha) \longrightarrow G, \quad (Y, t, X) \longmapsto \exp(Y) t \exp(X),$$

defined on points by the displayed formula, is an open immersion.

Suppose for the moment that the required exponential morphisms exist, and prove the other assertions. We first prove (ii). Both sides are of finite presentation and flat over S , so it is enough to work when S is the spectrum of an algebraically closed field (SGA 1, I 5.7 and VIII 5.5). Put $S = \text{Spec}(k)$, and choose

$$X \in \Gamma(S, \mathfrak{g}^\alpha)^\times, \quad Y \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times.$$

It is enough to prove that

$$\mathbb{G}_{a,k} \times_k T \times_k \mathbb{G}_{a,k} \longrightarrow G, \quad (y, t, x) \longmapsto \exp(yY) t \exp(xX),$$

is an open immersion. By 1.1 and 1.2, there exist $a, b \in k$ such that

$$\exp(yY) = p_{-\alpha}(ay), \quad \exp(xX) = p_\alpha(bx).$$

Since $\exp: W(\mathfrak{g}^{-\alpha}) \rightarrow G$ induces a monomorphism on Lie algebras, $a \neq 0$; likewise $b \neq 0$. The assertion is therefore reduced to 1.1.

The uniqueness of $\exp$ may be proved locally on S . We reduce to the case in which $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are free, and then apply 1.2 with $Q = T$ and $q = 1$.

It remains to prove existence. Faithfully flat descent, together with the uniqueness just proved, shows that it is enough to prove existence locally on S for the fpqc topology. The usual finite-presentation arguments reduce first to S noetherian and then to S noetherian local. We may therefore assume that

$$S = \text{Spec}(A),$$

where A is a complete noetherian local ring with algebraically closed residue field k . Let

$$p_0: \mathbb{G}_{a,k} \longrightarrow G_k$$

be a monomorphism of k -groups normalized by T_k , with multiplier $\alpha_0 = \alpha \otimes_A k$; such a morphism exists by 1.1. By 1.1 and 1.2, the associated morphism

$$T_k \cdot \alpha_0 \mathbb{G}_{a,k} \longrightarrow G_k$$

is an immersion and hence, in particular, a monomorphism. We provisionally admit the following two lemmas.

²It will be seen below, in Corollary 5.9, that $\exp$ is an isomorphism from $W(\mathfrak{g}^\alpha)$ onto a closed subgroup of G .

Lemma 1.6. *Let S be a scheme, G an S -group of finite presentation, T an S -torus, α a character of T that is nontrivial on every fiber, and $s_0 \in S$. Let*

$$f: T \cdot_{\alpha} \mathbb{G}_{a,S} \longrightarrow G$$

be an S -group morphism such that f_{s_0} is a monomorphism and the restriction of f to T is a monomorphism. There exists an open neighborhood U of s_0 such that $f|_U$ is a monomorphism.

Lemma 1.7. *Let A be a complete noetherian local ring with algebraically closed residue field k , let (G, T, α) be an A -elementary system, and let*

$$p_0: \mathbb{G}_{a,k} \longrightarrow G_k$$

be a morphism of k -groups normalized by T_k , with multiplier $\alpha \otimes_A k$. There exists a group morphism

$$p: \mathbb{G}_{a,A} \longrightarrow G$$

normalized by T , with multiplier α .

Let p be the morphism asserted by 1.7, and let

$$f: T \cdot_{\alpha} \mathbb{G}_{a,S} \longrightarrow G$$

be the corresponding morphism. It satisfies the hypotheses of 1.6, so it is a monomorphism; in particular, p is a monomorphism. The theorem now follows from Exposé XIX, 4.9.

Proof of 1.6. Let

$$\epsilon: S \longrightarrow T \cdot_{\alpha} \mathbb{G}_{a,S}$$

be the identity section. Since f is unramified at $\epsilon(s_0)$, it is unramified at $\epsilon(s)$ for all s in an open neighborhood U of s_0 . Hence $f|_U$ is unramified (Exposé X, 3.5),³ and its kernel $\text{Ker}(f)_U$ is unramified over U . To prove that $f|_U$ is a monomorphism, it is therefore enough⁴ to prove that $\text{Ker}(f)_U$ is radicial over U , which is a set-theoretic question. We are reduced to the following assertion.

Lemma 1.8. *Let k be an algebraically closed field, and let N be an invariant subgroup of $T \cdot_{\alpha} \mathbb{G}_{a,k}$, where α is a nontrivial character of the torus T . Suppose that N is étale over k and that $N \cap T = \{e\}$. Then $N = \{e\}$.*

We have

$$\text{int}(t')(t, x) = (t, \alpha(t')x).$$

If (t, x) is a point of N with $x \neq 0$, then (t, zx) is also a point of N for every $z \in k^{\times}$. Thus (t, x) is not isolated, contradicting the quasi-finiteness of N . Set-theoretically, therefore, $N \subset T$, and the hypothesis $N \cap T = \{e\}$ completes the proof.

Proof of 1.7. Let $\mathfrak{m}$ be the radical of A , and put

$$S_n = \text{Spec}(A/\mathfrak{m}^{n+1}), \quad n \geq 0.$$

We first show, by induction on n , that p_0 can be extended, for every n , to a morphism of S_n -groups

$$p_n: \mathbb{G}_{a,S_n} \longrightarrow G_{S_n},$$

³See also Exposé VI_B, 2.5.

⁴By EGA IV₄, 17.9.1.

normalized by T_{S_n} with multiplier α_{S_n} , in such a way that

$$p_{n+1} \times_{S_{n+1}} S_n = p_n.$$

Let $H = T \cdot_{\alpha} \mathbb{G}_{a,S}$. Denote by f_n the morphism $H_{S_n} \rightarrow G_{S_n}$ defined by p_n . We provisionally admit the following lemma.

Lemma 1.9. *Let (G, T, α) be a k -elementary system, where k is algebraically closed, and let*

$$p: \mathbb{G}_{a,k} \longrightarrow G$$

be a monomorphism normalized by T , with multiplier α . Then

$$H^2(T \cdot_{\alpha} \mathbb{G}_{a,k}, \mathfrak{g}) = 0.$$

Here $T \cdot_{\alpha} \mathbb{G}_{a,k}$ acts on $\mathfrak{g}$ through the morphism $T \cdot_{\alpha} \mathbb{G}_{a,k} \rightarrow G$ defined by p and through the adjoint representation of G .

By Exposé III, 2.8, f_n therefore extends to a morphism of S_{n+1} -groups

$$f'_{n+1}: H_{S_{n+1}} \longrightarrow G_{S_{n+1}}.$$

The morphism f'_{n+1} and the canonical immersion of $T_{S_{n+1}}$ into $G_{S_{n+1}}$ have the same restriction to T_{S_n} . By Exposé III, 2.5, there is an element $g \in G(S_{n+1})$ such that

$$g \times_{S_{n+1}} S_n = e$$

and such that

$$f_{n+1} = \text{int}(g) \circ f'_{n+1}$$

restricts to $T_{S_{n+1}}$ by the canonical immersion. Let p_{n+1} be the restriction of f_{n+1} to $\mathbb{G}_{a,S_{n+1}}$. It is normalized by $T_{S_{n+1}}$, has multiplier $\alpha_{S_{n+1}}$, and extends p_n .

We have thus constructed a coherent system (f_n) ; it remains to algebraize it. For this we use:

Lemma 1.10. *Let A be a complete noetherian local ring, $\mathfrak{m}$ its maximal ideal,*

$$S = \text{Spec}(A), \quad S_n = \text{Spec}(A/\mathfrak{m}^{n+1}),$$

let T be an S -torus, let α be a nonzero character of T , and let X be an affine S -scheme on which T acts. Let T act on $\mathbb{G}_{a,S}$ through α . Let $q \geq 0$, and let $(f_n)_{n \geq 0}$ be a coherent system of morphisms

$$f_n: \mathbb{G}_{a,S_n}^q \longrightarrow X_{S_n}$$

of objects with operators T_{S_n} . There exists a unique morphism of objects with operators T

$$f: \mathbb{G}_{a,S}^q \longrightarrow X$$

that induces the f_n (compare Exposé IX, 7.1).

Corollary 1.11. *If X is a group with operator group T , and if the f_n are group morphisms, then f is also a group morphism.*

Indeed, apply the uniqueness assertion of the lemma to the two morphisms $\mathbb{G}_{a,S}^{2q} \rightarrow X$ deduced from f in the usual way.

Proof of 1.10. Suppose that T is split, as it is in the application of 1.10 to the proof of 1.5. By Exposé I, 4.7.3.1, the functor

$$X \mapsto \mathcal{A}(X)$$

defines an equivalence between the category of affine S -schemes equipped with an action of T and the category opposite to that of M -graded S -algebras, where

$$M = \text{Hom}_{S\text{-gr.}}(T, \mathbb{G}_{m,S}).$$

Thus there are gradings

$$B = \mathcal{A}(X) = \bigoplus_{m \in M} B_m, \quad C = \mathcal{A}(\mathbb{G}_{a,S}^q) = \bigoplus_{m \in M} C_m.$$

Each C_m is visibly a finite free A -module. Indeed, $C_m = 0$ if m is not a multiple of α , whereas, if $m = d\alpha$, then C_m is isomorphic to the A -module of homogeneous polynomials of degree d in q variables. Put

$$\begin{aligned} \hat{B}_m &= \varprojlim_n B_m \otimes_A (A/\mathfrak{m}^{n+1}), \\ \hat{C}_m &= \varprojlim_n C_m \otimes_A (A/\mathfrak{m}^{n+1}), \\ \hat{B} &= \bigoplus_{m \in M} \hat{B}_m, & \hat{C} &= \bigoplus_{m \in M} \hat{C}_m. \end{aligned}$$

There are canonical morphisms of M -graded algebras

$$g_B: B \longrightarrow \hat{B}, \quad g_C: C \longrightarrow \hat{C}.$$

The preceding observation shows that g_C is an isomorphism. Giving a coherent system (f_n) as in the statement is equivalent to giving a morphism of graded A -algebras

$$\hat{F}: \hat{B} \longrightarrow \hat{C}.$$

Finding a morphism f as in the statement is equivalent to finding a morphism of graded A -algebras $F: B \rightarrow C$ making the diagram

$$\begin{array}{ccc} B & \xrightarrow{F} & C \\ g_B \downarrow & & \downarrow g_C \\ \hat{B} & \xrightarrow{\hat{F}} & \hat{C} \end{array}$$

commute. Since g_C is an isomorphism, the existence and uniqueness of F are immediate. This proves 1.10.

To complete the proof of 1.5, it remains only to prove 1.9.

1.12. Proof of 1.9. We have

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha}.$$

As explained in 1.9, regard $\mathfrak{g}$ as a $T \cdot_\alpha \mathbb{G}_{a,k}$ -module. Clearly $\mathfrak{t} \oplus \mathfrak{g}^\alpha$ is a submodule of $\mathfrak{g}$, and the quotient is isomorphic to $\mathfrak{g}^{-\alpha}$ as a k -vector space and even as a T -module. The group $\mathbb{G}_{a,k}$ acts trivially on this one-dimensional quotient, since every group morphism $\mathbb{G}_{a,k} \rightarrow \mathbb{G}_{m,k}$ is trivial. Likewise, $\mathfrak{g}^\alpha$ is a submodule of $\mathfrak{t} \oplus \mathfrak{g}^\alpha$; the quotient is isomorphic to $\mathfrak{t}$ as a T -module, and $\mathbb{G}_{a,k}$ acts trivially on it. In summary:

Lemma 1.13. *Under the hypotheses of 1.9, $\mathfrak{g}$, as a $T \cdot_\alpha \mathbb{G}_{a,k}$ -module, has a composition series whose successive quotients are*

$$\mathfrak{g}^{-\alpha}, \quad \mathfrak{t}, \quad \mathfrak{g}^\alpha,$$

regarded as $T \cdot_\alpha \mathbb{G}_{a,k}$ -modules through the projection $T \cdot_\alpha \mathbb{G}_{a,k} \rightarrow T$.

We are thus reduced to computing the cohomology of $T \cdot_\alpha \mathbb{G}_{a,k}$ acting on $W(k)$ through the projection $T \cdot_\alpha \mathbb{G}_{a,k} \rightarrow T$ and a character β of T (here $\beta = 0, \alpha$, or $-\alpha$).⁵ Write $k[x_1, \dots, x_n]$ for the polynomial algebra over k in n variables, and $k_q[x_1, \dots, x_n]$ for its subspace of homogeneous polynomials of degree q .

Lemma 1.14. *With the preceding notation,*

$$H^n(T \cdot_\alpha \mathbb{G}_{a,k}, k) = H^n(C_{\alpha, \beta}^*),$$

where the complex $C_{\alpha, \beta}^$ is defined by*

$$C_{\alpha, \beta}^n = \begin{cases} k_q[x_1, \dots, x_n], & \text{if } \beta = q\alpha, \text{ with } q \in \mathbb{N}^*, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$\begin{aligned} (\delta f)(x_1, x_2, \dots, x_{n+1}) &= f(x_2, \dots, x_{n+1}) \\ &\quad + \sum_{i=1}^n (-1)^i f(x_1, \dots, x_i x_{i+1}, \dots, x_{n+1}) \\ &\quad + (-1)^{n+1} f(x_1, \dots, x_n). \end{aligned}$$

Indeed, by Exposé I, 5.3.2, the functor $M \mapsto H^0(T, M)$ is exact on the category of T -modules, and the $H^q(T, -)$ vanish. It follows, as in the usual theory of group cohomology, that $H^n(T \cdot_\alpha \mathbb{G}_{a,k}, k)$ may be computed as the n -th cohomology group of the complex of cochains of $\mathbb{G}_{a,k}$ in k that are invariant under T , that is, those satisfying

$$f(\alpha(t)x_1, \dots, \alpha(t)x_n) = \beta(t)f(x_1, \dots, x_n).$$

This is precisely the complex just displayed.

To prove 1.9, it is therefore enough to show that $H^2(C_{\alpha, \beta}^*) = 0$ for $\beta = 0, \alpha, -\alpha$, which is immediate.

Remark 1.15. The groups $H^n(C_{\alpha, \beta}^*)$ can be computed explicitly when $\beta = q\alpha$; see M. Lazard, *Lois de groupes et analyseurs*, Annales E.N.S., 1955. In particular,

$$H^n(C_{\alpha, q\alpha}^*) = 0 \quad \text{for } n > q.$$

⁵The editors added the sentence that follows.

Notations 1.16. The image of the canonical immersion

$$W(\mathfrak{g}^{-\alpha}) \times_S T \times_S W(\mathfrak{g}^{\alpha}) \longrightarrow G$$

will be denoted by Ω . It is an open subset of G containing the identity section. The images of

$$W(\mathfrak{g}^{-\alpha}), \quad W(\mathfrak{g}^{\alpha}), \quad W(\mathfrak{g}^{-\alpha}) \times_S T, \quad T \times_S W(\mathfrak{g}^{\alpha})$$

will be denoted, respectively,⁶

$$U_{-\alpha}, \quad U_{\alpha}, \quad U_{-\alpha} \cdot T, \quad T \cdot U_{\alpha}.$$

Then U_{α} (respectively $U_{-\alpha}$) is a subgroup of G , canonically endowed with a vector-bundle structure, and

$$\text{int}(t)(x) = x^{\alpha(t)} \quad (\text{respectively } x^{-\alpha(t)})$$

for every $S' \rightarrow S$, $t \in T(S')$, and $x \in U_{\alpha}(S')$ (respectively $x \in U_{-\alpha}(S')$).

There are canonical isomorphisms

$$T \cdot U_{\alpha} \simeq T \cdot_{\alpha} U_{\alpha}, \quad T \cdot U_{-\alpha} \simeq T \cdot_{-\alpha} U_{-\alpha}.$$

The open subset Ω is stable under $\text{int}(T)$, and

$$\text{int}(t')(y \cdot t \cdot x) = y^{-\alpha(t')} \cdot t \cdot x^{\alpha(t')}.$$

Corollary 1.17. *We have*

$$\text{Lie}(U_{\alpha}/S) = \mathfrak{g}^{\alpha}, \quad \text{Lie}(U_{-\alpha}/S) = \mathfrak{g}^{-\alpha}.$$

The isomorphisms

$$W(\mathfrak{g}^{\alpha}) \xrightarrow[\sim]{\exp} U_{\alpha}, \quad W(\mathfrak{g}^{-\alpha}) \xrightarrow[\sim]{\exp} U_{-\alpha}$$

are those of Exposé XIX, 4.2.

Corollary 1.18. *The open subset Ω is relatively schematically dense in G (cf. Exposé XVIII, §1).*

This follows immediately from Exposé XVIII, 1.3.⁷

Corollary 1.19. *The center of G is*

$$\text{Centr}(G) = \text{Ker}(\alpha).$$

It is therefore a closed subgroup of G , of multiplicative type and of finite type.

The second assertion follows from the first by Exposé IX, 2.7. Let us prove the first. The inner automorphism defined by a section of $\text{Ker}(\alpha)$ acts trivially on Ω , by the last formula of 1.16, and hence on G , by 1.18. Conversely, if $g \in G(S)$ centralizes G , then it centralizes T and U_{α} . It is therefore a section of T , by Exposé XIX, 2.8, and annihilates α . Since the same argument applies after every base change, we indeed have $\text{Centr}(G) = \text{Ker}(\alpha)$.

⁶The editors replaced $P_{-\alpha}$ and P_{α} by $U_{-\alpha}$ and U_{α} .

⁷See also EGA IV₃, 11.10.10.

Corollary 1.20. *A monomorphism*

$$p_\alpha: \mathbb{G}_{a,S} \longrightarrow G$$

normalized by T , with multiplier α , exists if and only if the $\mathcal{O}_S$ -module $\mathfrak{g}^\alpha$ is free. More precisely, the assignments

$$X_\alpha \longmapsto (x \longmapsto \exp(xX_\alpha)), \quad p_\alpha \longmapsto \text{Lie}(p_\alpha)$$

give a bijection between $\Gamma(S, \mathfrak{g}^\alpha)^\times$ and the set of such monomorphisms p_α , which is also the set of isomorphisms of vector group schemes

$$\mathbb{G}_{a,S} \xrightarrow{\sim} U_\alpha.$$

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Corollary 1.21. *The subgroups U_α and $U_{-\alpha}$ of G commute on no fiber.*

Indeed, if $(U_\alpha)_s$ and $(U_{-\alpha})_s$ commute, then Ω_s is a subgroup of G_s , and hence $\Omega_s = G_s$.⁹ It follows that G_s is solvable, contrary to the hypothesis that G_s is reductive of semisimple rank 1.

2 Structure of elementary systems

Theorem 2.1. *Let S be a scheme and (G, T, α) an S -elementary system. There exist a morphism of $\mathcal{O}_S$ -modules*

$$\mathfrak{g}^\alpha \otimes_{\mathcal{O}_S} \mathfrak{g}^{-\alpha} \longrightarrow \mathcal{O}_S, \quad (X, Y) \longmapsto \langle X, Y \rangle,$$

and a morphism of S -groups

$$\alpha^*: \mathbb{G}_{m,S} \longrightarrow T$$

such that, for every $S' \rightarrow S$ and all

$$X \in \Gamma(S', \mathfrak{g}^\alpha \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}), \quad Y \in \Gamma(S', \mathfrak{g}^{-\alpha} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}),$$

one has

$$\exp(X) \cdot \exp(Y) \in \Omega(S') \iff 1 + \langle X, Y \rangle \in \mathbb{G}_m(S').$$

Under these conditions,

$$(F) \quad \begin{aligned} \exp(X) \cdot \exp(Y) &= \exp\left(\frac{Y}{1 + \langle X, Y \rangle}\right) \alpha^*(1 + \langle X, Y \rangle) \\ &\quad \exp\left(\frac{X}{1 + \langle X, Y \rangle}\right). \end{aligned}$$

Moreover, the morphisms $(X, Y) \mapsto \langle X, Y \rangle$ and α^ are uniquely determined. The first is an isomorphism, and therefore places $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ in duality; furthermore,*

$$\alpha \circ \alpha^* = 2,$$

where 2 denotes squaring on $\mathbb{G}_{m,S}$.

⁸Indeed, $\text{Lie}(\mathbb{G}_{a,S}) = \mathcal{O}_S$, and $\text{Lie}(p_\alpha)$ is an element of $\text{Hom}_{\mathcal{O}_S}(\mathcal{O}_S, \mathfrak{g}^\alpha) = \Gamma(S, \mathfrak{g}^\alpha)$.

⁹By 1.18.

In view of the uniqueness assertions, it is enough to prove the theorem locally on S . We may therefore assume that $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are free on S . Choose

$$X \in \Gamma(S, \mathfrak{g}^\alpha)^\times, \quad Y \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times,$$

and put

$$p_\alpha(x) = \exp(xX), \quad p_{-\alpha}(y) = \exp(yY)$$

for $x, y \in \mathbb{G}_a(S')$, $S' \rightarrow S$. By 1.5 and 1.21, it is enough to prove the following proposition.

Proposition 2.2. *Let S be a scheme, G an S -group, T a torus of G , and α a character of T that is nontrivial on every fiber. Let*

$$p_\alpha: \mathbb{G}_{a,S} \longrightarrow G, \quad p_{-\alpha}: \mathbb{G}_{a,S} \longrightarrow G$$

be group monomorphisms normalized by T , with multipliers α and $-\alpha$, respectively. Suppose that:

(i) *the morphism*

$$\mathbb{G}_{a,S} \times_S T \times_S \mathbb{G}_{a,S} \longrightarrow G, \quad (y, t, x) \longmapsto p_{-\alpha}(y) t p_\alpha(x),$$

is an open immersion; denote its image by Ω ;

(ii) *for every $s \in S$, the groups*

$$(p_\alpha)_s(\mathbb{G}_{a,\kappa(s)}) \quad \text{and} \quad (p_{-\alpha})_s(\mathbb{G}_{a,\kappa(s)})$$

do not commute.

Then there exist unique elements

$$a \in \mathbb{G}_a(S), \quad \alpha^* \in \text{Hom}_{S\text{-gr.}}(\mathbb{G}_{m,S}, T)$$

with the following properties: for every $S' \rightarrow S$ and all $x, y \in \mathbb{G}_a(S')$,

$$p_\alpha(x) p_{-\alpha}(y) \in \Omega(S') \iff 1 + axy \in \mathbb{G}_m(S'),$$

and, under this condition,

$$p_\alpha(x) p_{-\alpha}(y) = p_{-\alpha} \left(\frac{y}{1 + axy} \right) \alpha^*(1 + axy) p_\alpha \left(\frac{x}{1 + axy} \right).$$

Moreover, a is invertible, that is, $a \in \mathbb{G}_m(S)$, and $\alpha \circ \alpha^ = 2$.*

Proof. A) Consider the morphism

$$\mathbb{G}_{a,S}^2 \longrightarrow G, \quad (x, y) \longmapsto p_\alpha(x) p_{-\alpha}(y).$$

Let U be the inverse image of Ω . It is an open subset of $\mathbb{G}_{a,S}^2$ containing $0 \times_S \mathbb{G}_{a,S}$ and $\mathbb{G}_{a,S} \times_S 0$. There are therefore unique morphisms of S -schemes

$$A: U \longrightarrow \mathbb{G}_{a,S}, \quad C: U \longrightarrow \mathbb{G}_{a,S}, \quad B: U \longrightarrow T$$

satisfying the set-theoretic identity

$$p_\alpha(u) p_{-\alpha}(v) = p_{-\alpha}(A(u, v)) B(u, v) p_\alpha(C(u, v)).$$

Immediately,

$$\begin{aligned} A(0, v) &= v, & A(u, 0) &= 0, & C(u, 0) &= u, & C(0, v) &= 0, \\ B(u, 0) &= B(0, v) = e. \end{aligned}$$

Let S' be a separated S -scheme and let $t \in T(S')$. Since $\Omega_{S'}$ is stable under $\text{int}(t)$, the last formula of 1.16 shows that $U_{S'}$ is stable under the automorphism

$$(x, y) \mapsto (\alpha(t)x, \alpha(t)^{-1}y)$$

of $\mathbb{G}_{a, S'}^2$, and

$$\begin{aligned} A(\alpha(t)u, \alpha(t)^{-1}v) &= \alpha(t)^{-1}A(u, v), \\ C(\alpha(t)u, \alpha(t)^{-1}v) &= \alpha(t)C(u, v), \\ B(\alpha(t)u, \alpha(t)^{-1}v) &= B(u, v). \end{aligned}$$

Because α is faithfully flat, it follows that for every $S' \rightarrow S$ and every $z \in \mathbb{G}_m(S')$, the open subset $U_{S'}$ is stable under $(x, y) \mapsto (zx, z^{-1}y)$, and

$$\begin{aligned} A(zu, z^{-1}v) &= z^{-1}A(u, v), \\ C(zu, z^{-1}v) &= zC(u, v), \\ B(zu, z^{-1}v) &= B(u, v). \end{aligned}$$

Suppose first that v is invertible. Taking $z = v$, we see that if (u, v) is a section of U , then $(uv, 1)$ is also a section of U , and

$$A(uv, 1) = v^{-1}A(u, v), \quad B(uv, 1) = B(u, v).$$

Let V be the open subset of $\mathbb{G}_{a, S}^2$ defined by¹⁰

$$(u, v) \in V(S') \iff (u, v), (uv, 1), \text{ and } (1, uv) \text{ belong to } U(S').$$

Since U contains $0 \times_S \mathbb{G}_{a, S}$ and $\mathbb{G}_{a, S} \times_S 0$, the open subset V is a neighborhood of the zero section of $\mathbb{G}_{a, S}^2$. The preceding calculation shows that, on $V \cap (\mathbb{G}_{a, S} \times_S \mathbb{G}_{m, S})$, the two morphisms

$$(u, v) \mapsto A(u, v), \quad (u, v) \mapsto vA(uv, 1)$$

coincide, as do

$$(u, v) \mapsto B(u, v), \quad (u, v) \mapsto B(uv, 1).$$

Since $\mathbb{G}_{a, S} \times_S \mathbb{G}_{m, S}$ is schematically dense in $\mathbb{G}_{a, S}^2$, these morphisms coincide on V .

We have $A(0, 1) = 1$. Hence there is an open subset W_1 of $\mathbb{G}_{a, S}$, containing the zero section, such that $A(x, 1)$ is invertible for every section x of W_1 . Put

$$A(x, 1)^{-1} = F(x).$$

If $(u, v) \in V(S')$ ¹¹ and $uv \in W_1(S')$, then

$$A(u, v) = vA(uv, 1) = vF(uv)^{-1}.$$

¹⁰The editors added the condition “ $(1, uv) \in U(S')$ ”.

¹¹Here and below, the editors replaced $U(S')$ by $V(S')$.

The same argument applied to C gives an open subset W_2 of $\mathbb{G}_{a,S}$, containing the zero section, and an element $E \in \mathcal{O}(W_2)^{\times 12}$ such that

$$C(u, v) = uC(1, uv) = uE(uv)^{-1}$$

whenever $(u, v) \in V(S')$ and $uv \in W_2(S')$.

Consequently, with $W = W_1 \cap W_2$, there are S -morphisms

$$\begin{aligned} F: W &\rightarrow \mathbb{G}_{m,S}, & F(0) &= 1, \\ H: W &\rightarrow T, & H(0) &= e, \\ E: W &\rightarrow \mathbb{G}_{m,S}, & E(0) &= 1, \end{aligned}$$

such that, if $(u, v) \in V(S')$ and $uv \in W(S')$, then

$$(+) \quad p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}(vF(uv)^{-1})H(uv)p_\alpha(uE(uv)^{-1}).$$

B) We now use the associativity of G to write

$$p_\alpha(u)p_{-\alpha}(v)p_{-\alpha}(w) = p_\alpha(u)p_{-\alpha}(v+w).$$

There is an open subset L of $\mathbb{G}_{a,S}^3$, containing the zero section, such that $(u, v, w) \in L(S')$ is equivalent to

$$\begin{aligned} (u, v) &\in V(S'), & (uE(uv)^{-1}, w) &\in V(S'), & (u, v+w) &\in V(S'), \\ uv &\in W(S'), & uwE(uv)^{-1} &\in W(S'), & u(v+w) &\in W(S'). \end{aligned}$$

Using formula (+), we immediately obtain, for $(u, v, w) \in L(S')$,

$$(1) \quad E(uv+uw) = E(uwE(uv)^{-1})E(uv),$$

$$(2) \quad H(uv+uw) = H(uwE(uv)^{-1})H(uv),$$

and

$$(3) \quad \begin{aligned} (v+w)F(uv+uw)^{-1} &= \alpha(H(uv)^{-1})wF(uwE(uv)^{-1})^{-1} \\ &\quad + vF(uv)^{-1}. \end{aligned}$$

It follows immediately from the definition of L that $(1, 0, 0) \in L(S)$. Consider therefore

$$L \cap_T (1 \times_S \mathbb{G}_{a,S} \times_S \mathbb{G}_{a,S}) = 1 \times_S M.$$

The scheme M is an open subset of $\mathbb{G}_{a,S}^2$ containing the section $(0, 0)$. For $(v, w) \in M(S')$, the elements

$$v, \quad wE(v)^{-1}, \quad v+w$$

belong to $W(S')$, and

$$(1') \quad E(v+w) = E(wE(v)^{-1})E(v),$$

$$(2') \quad H(v+w) = H(wE(v)^{-1})H(v),$$

¹²The editors use E for the element denoted G in the original, since G already denotes the S -group under consideration.

$$(3') \quad (v+w)F(v+w)^{-1} = \alpha(H(v))^{-1}wF(wE(v)^{-1})^{-1} \\ + vF(v)^{-1}.$$

Finally, consider the morphism from M to $\mathbb{G}_{a,S}^2$ defined set-theoretically by

$$(v, w) \mapsto (v, wE(v)^{-1}).$$

¹³ It preserves the section $(0, 0)$ and induces an isomorphism of M onto an open subset N of $\mathbb{G}_{a,S}^2$ containing the zero section; the inverse isomorphism is

$$(x, y) \mapsto (x, yE(x)).$$

¹⁴ We have therefore proved the following assertion:

There is an open subset N of $\mathbb{G}_{a,S}^2$, containing the zero section, such that, if $(x, y) \in N(S')$, then x , y , and $x + yE(x)$ ¹⁵ belong to $W(S')$, and

$$(1'') \quad E(x + yE(x)) = E(x)E(y),$$

$$(2'') \quad H(x + yE(x)) = H(x)H(y),$$

$$(3'') \quad (x + yE(x))F(x + yE(x))^{-1} = xF(x)^{-1} \\ + \alpha(H(x))^{-1}yE(x)F(y)^{-1}.$$

C) Arguing in the same way with left associativity, we obtain the following assertion.¹⁶

There is an open subset N' of $\mathbb{G}_{a,S}^2$, containing the zero section, such that, if $(x, y) \in N'(S')$, then x , y , and $x + yF(x)$ ¹⁷ belong to $W(S')$, and

$$(4'') \quad F(x + yF(x)) = F(x)F(y),$$

$$(5'') \quad H(x + yF(x)) = H(x)H(y),$$

$$(6'') \quad (x + yF(x))E(x + yF(x))^{-1} = xE(x)^{-1} \\ + \alpha(H(x))^{-1}yF(x)E(y)^{-1}.$$

We are thus led to solve the “functional equation” (1'').

¹³The editors corrected what follows.

¹⁴Thus the editors made the change of variables $x = v$, $y = wE(v)^{-1}$, or equivalently $v = x$, $w = yE(x)$.

¹⁵The editors corrected $yE(x)$ to $x + yE(x)$.

¹⁶That is, one writes the equalities resulting from $p_\alpha(t)p_\alpha(u)p_{-\alpha}(v) = p_\alpha(t+u)p_{-\alpha}(v)$, and sets $v = 1$, $x = u$, $t = yF(u)$, or equivalently $y = tF(u)^{-1}$.

¹⁷The editors corrected $yF(x)$ to $x + yF(x)$.

Lemma 2.3. *Let S be a scheme, let W be an open subset of $\mathbb{G}_{a,S}$ containing the zero section, and let*

$$F: W \longrightarrow \mathbb{G}_{m,S}$$

be an S -morphism. Suppose that $F(0) = 1$ and that there is an open subset N of $\mathbb{G}_{a,S}^2$, containing the zero section, such that, for $(x, y) \in N(S')$, the elements x , y , and $x + yF(x)$ belong to $W(S')$, and

$$(\dagger) \quad F(x + yF(x)) = F(x)F(y).$$

(i) *If $S = \operatorname{Spec}(k)$ for a field k , there is an $a \in k$ such that $F(x) = 1 + ax$.*

(ii) *If $a = F'(0) \in \Gamma(S, \mathcal{O}_S)$ is invertible, then $F(x) = 1 + ax$.*

Proof. Under the hypotheses, we may differentiate the given equation at $x = 0$, respectively at $y = 0$, and obtain

$$(*) \quad F'(y)(1 + yF'(0)) = F'(0)F(y) \quad \text{for } (0, y) \in N(S'),$$

respectively

$$F'(x)F(x) = F(x)F'(0) \quad \text{for } (x, 0) \in N(S').$$

Since F takes its values in $\mathbb{G}_m$, the second relation gives

$$(*)' \quad F'(x) = F'(0) \quad \text{for } (x, 0) \in N(S').$$

The first relation therefore gives

$$F'(0)(1 + yF'(0)) = F'(0)F(y) \quad \text{for } (y, 0), (0, y) \in N(S').$$

If $a = F'(0)$ is invertible, it follows that

$$F(y) = 1 + ay$$

for y a section of an open subset of W containing the zero section, hence schematically dense in W . This proves (ii), and also proves (i) when $F'(0) \neq 0$.

Suppose $F'(0) = 0$. Then $(*)'$ shows that $F'(x) = 0$ for x near 0, and hence $F' = 0$ by schematic density. If k has characteristic zero, F is a rational function with zero derivative, and is therefore constant and equal to $F(0) = 1$.

Suppose now that k has characteristic p , and that F is not constant.¹⁸ There are an integer $n > 0$ and a rational function $F_1 \in k(X)$ such that $F_1'(X) \neq 0$ and

$$F(X) = F_1(X^{p^n}) = F_1(X)^{p^n}.$$

Substitution in the functional equation gives

$$(\dagger_1) \quad F_1(x + yF_1(x)^{p^n}) = F_1(x)F_1(y).$$

Differentiating at $x = 0$ gives

$$(*_1) \quad F_1'(y) = F_1'(0)F_1(y),$$

whereas differentiation of $(\dagger_1)$ at $y = 0$ gives

$$(*'_1) \quad F_1'(x)F_1(x)^{p^n} = F_1(x)F_1'(0).$$

¹⁸The editors corrected what follows.

By hypothesis, $F'_1(X)$ is invertible in $k(X)$; these two equalities therefore imply

$$F_1(X)^{p^n} = 1.$$

Thus F_1 is constant, contrary to the choice of F_1 . Consequently F is constant and equal to $1 = F(0)$. $\square$

D) Suppose first that S is the spectrum of a field. If $F'(0) = 0$, then $F = 1$. Formula (5'') then gives

$$H(x + y) = H(x)H(y),$$

which shows that H extends to a group morphism

$$\mathbb{G}_{a,S} \longrightarrow T$$

(Exposé XVIII, 2.3), necessarily the constant morphism with value e . On the other hand, Lemma 2.3 also gives

$$E(x) = 1 + bx$$

for some $b \in k$. Formula (6'') therefore gives, for $(x, y) \in N(S')$,

$$(x + y)E(x + y)^{-1} = xE(x)^{-1} + yE(y)^{-1}.$$

Thus,¹⁹ again by Exposé XVIII, 2.3, the map

$$x \longmapsto xE(x)^{-1}$$

extends to a morphism of k -groups $\mathbb{G}_{a,k} \rightarrow \mathbb{G}_{a,k}$. Consequently

$$\frac{x}{1 + bx} = cx$$

for some $c \in k$, whence $b = 0$ and $c = 1$.

Thus F, H, E are constant with values $(1, e, 1)$ in a neighborhood of the zero section, and hence everywhere. Formula (+) then shows that U_α and $U_{-\alpha}$ commute, contrary to hypothesis (ii).

For arbitrary S , we have therefore proved that $F'(0)$ is nonzero on every fiber, and hence invertible. The same is plainly true of $E'(0)$. Lemma 2.3 now shows that there are $a, b \in \mathbb{G}_m(S)$ such that

$$(\diamond_1) \quad F(x) = 1 + ax, \quad E(x) = 1 + bx \quad \text{for } x \in W(S').$$

E) The rest is now easy. Substitution of the preceding results in (3'') gives

$$y \alpha(H(x))(1 + ay) = y(1 + ax + ay(1 + bx))(1 + ax).$$

This formula holds for every section (x, y) of N . Since $\mathbb{G}_{a,S} \times_S \mathbb{G}_{m,S}$ is schematically dense in $\mathbb{G}_{a,S}^2$, it follows that

$$(1 + ay)\alpha(H(x)) = (1 + ax + ay(1 + bx))(1 + ax).$$

Setting $y = 0$ gives

$$\alpha(H(x)) = (1 + ax)^2.$$

¹⁹The editors added the sentence that follows. One may also see by direct calculation that the preceding equality implies $0 = xyb(2 + (x + y)b)$, hence $0 = b(2 + (x + y)b)$, and finally $b = 0$.

Substitution in the preceding equality gives²⁰

$$a^2xy = abxy.$$

Since $\mathbb{G}_{m,S}$ is schematically dense in $\mathbb{G}_{a,S}$, we obtain $a^2 = ab$. As a is invertible,

$$(\diamond_2) \quad a = b, \quad \alpha(H(x)) = (1 + ax)^2.$$

Because a is invertible, the map $x \mapsto 1 + ax$ is an automorphism of $\mathbb{G}_{a,S}$. There are therefore an open subset W' of $\mathbb{G}_{a,S}$, containing the section 1, and a morphism

$$P: W' \longrightarrow T$$

such that

$$P(1 + ax) = H(x).$$

²¹ Substitution in $(2')$ gives immediately, for $(x, y) \in N(S')$,

$$P(1 + ax + ay) = P\left(\frac{1 + ax + ay}{1 + ax}\right) P(1 + ax).$$

Thus there is an open neighborhood of 1 in $\mathbb{G}_{m,S}$ on which

$$P(x)P(y) = P(xy).$$

By Exposé XVIII, 2.3, there is a group morphism

$$(\diamond_3) \quad \alpha^*: \mathbb{G}_{m,S} \longrightarrow T$$

extending P . Since $\alpha(H(x)) = (1 + ax)^2$ near the zero section, we have $\alpha(\alpha^*(z)) = z^2$ near the section 1, and hence

$$(\diamond_4) \quad \alpha \circ \alpha^* = 2.$$

F)²² Combining $(+)$ and $(\diamond_1)$ – $(\diamond_4)$, we see that there are

$$a \in \mathbb{G}_m(S), \quad \alpha^* \in \text{Hom}_{S\text{-gr.}}(\mathbb{G}_{m,S}, T)$$

such that $\alpha \circ \alpha^* = 2$, and such that, whenever $(u, v) \in V(S')$ and $uv \in W(S')$, the element $1 + auv$ is invertible and

$$p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}\left(\frac{v}{1 + auv}\right) \alpha^*(1 + auv) p_\alpha\left(\frac{u}{1 + auv}\right).$$

Let V' be the open subset of $\mathbb{G}_{a,S}^2$ defined by the invertibility of $1 + auv$; thus

$$V' = (\mathbb{G}_{a,S}^2)_f, \quad f(u, v) = 1 + auv.$$

The two sides of the preceding formula define morphisms $V' \rightarrow G$ which agree in a neighborhood of the zero section, and hence agree on V' . Thus the formula holds for every

²⁰In the three preceding equalities, the editors corrected the original by replacing the factor $1 + bx$ on the right by $1 + ax$. Substitution of the third equality in the second, and use of the invertibility of $1 + ax$, gives precisely $a^2xy = abxy$.

²¹This is the change of variables $x' = 1 + ax$, or $x = (x' - 1)/a$.

²²The editors slightly modified what follows, since an open subset V had already been introduced in part A.

section (u, v) of V' , and in particular $V' \subset U$, where U is the open subset introduced at the beginning of part A.

We prove that $U = V'$. Returning to the notation of part A, there is a morphism

$$A: U \longrightarrow \mathbb{G}_{a,S}$$

whose restriction to V' is

$$A(u, v) = v(1 + auv)^{-1}.$$

The equality $U = V'$ is a set-theoretic question and may therefore be checked when $S = \text{Spec}(k)$ for a field k . It then reduces to the evident assertion that the domain of definition of the rational map

$$\mathbb{G}_{a,k}^2 \dashrightarrow \mathbb{G}_{a,k}, \quad (X, Y) \longmapsto \frac{Y}{1 + aXY},$$

is the open subset defined by $1 + aXY$.

G) We have proved the existence of a and α^* , as well as the two asserted supplementary properties. It remains to prove uniqueness. Let a' and $\alpha^{*'}$ also satisfy the required conditions. For $u, v \in \mathbb{G}_a(S')$, we immediately have

$$\begin{aligned} 1 + auv \text{ invertible} &\implies 1 + a'uv \text{ invertible}, \\ \frac{v}{1 + auv} &= \frac{v}{1 + a'uv}. \end{aligned}$$

Hence, for every $u \in \mathbb{G}_a(S')$,

$$1 + au \text{ invertible} \implies 1 + au = 1 + a'u,$$

which proves $a = a'$. With the same notation,

$$1 + au \text{ invertible} \implies \alpha^*(1 + au) = \alpha^{*'}(1 + au),$$

and therefore $\alpha^* = \alpha^{*'}$. □

Corollary 2.4. *Let*

$$\exp(Y)t\exp(X) \quad \text{and} \quad \exp(Y')t'\exp(X')$$

be two elements of $\Omega(S')$. Their product belongs to $\Omega(S')$ if and only if

$$u = 1 + \langle X, Y' \rangle$$

is invertible; in that case,

$$\begin{aligned} \text{(F')} \quad &\exp(Y)t\exp(X) \cdot \exp(Y')t'\exp(X') \\ &= \exp(Y + u^{-1}\alpha(t)^{-1}Y') \, tt'\alpha^*(u) \exp(u^{-1}\alpha(t')^{-1}X + X'). \end{aligned}$$

Remark 2.5. *Formula (F) of Theorem 2.1 may also be written without using the morphisms $\exp$. Transporting the duality $\mathfrak{g}_\alpha \otimes \mathfrak{g}_{-\alpha} \rightarrow \mathcal{O}_S$ through those morphisms gives a canonical pairing of vector bundles*

$$U_\alpha \times_S U_{-\alpha} \longrightarrow \mathbb{G}_{a,S},$$

which we continue to denote by $(x, y) \mapsto \langle x, y \rangle$. Thus

$$\langle \exp X, \exp Y \rangle = \langle X, Y \rangle.$$

If $x \in U_\alpha(S')$, $y \in U_{-\alpha}(S')$, and $1 + \langle x, y \rangle \in \mathbb{G}_m(S')$, then

$$x \cdot y = y^{(1 + \langle x, y \rangle)^{-1}} \alpha^*(1 + \langle x, y \rangle) x^{(1 + \langle x, y \rangle)^{-1}}.$$

Corollary 2.6. *The pairing*

$$W(\mathfrak{g}_\alpha) \times_S W(\mathfrak{g}_{-\alpha}) \longrightarrow \mathbb{G}_{a,S}$$

induces a pairing of principal $\mathbb{G}_{m,S}$ -bundles

$$W(\mathfrak{g}_\alpha)^\times \times_S W(\mathfrak{g}_{-\alpha})^\times \longrightarrow \mathbb{G}_{m,S}.$$

We denote this pairing by $(X, Y) \mapsto \langle X, Y \rangle$, or simply $(X, Y) \mapsto XY$.

For every $X \in \Gamma(S, \mathfrak{g}_\alpha)^\times$, there is therefore a unique

$$X^{-1} \in \Gamma(S, \mathfrak{g}_{-\alpha})^\times$$

such that $XX^{-1} = 1$. Moreover,

$$(zX)^{-1} = z^{-1}X^{-1}.$$

The morphism

$$s: W(\mathfrak{g}_\alpha)^\times \longrightarrow W(\mathfrak{g}_{-\alpha})^\times$$

so defined is an isomorphism of schemes, compatible with the isomorphism $s: z \mapsto z^{-1}$ of their operator groups.

Definition 2.6.1. *The elements X and $s(X) = X^{-1}$ are said to be paired.*

Apply Corollary 2.4 with $Y = 0 = X'$ and

$$Y' = aX^{-1}, \quad a \in \mathcal{O}_S(S).$$

Then $u = 1 + a$, and

$$u^{-1}Y' = u^{-1}(u - 1)X^{-1} = (1 - u^{-1})X^{-1}.$$

Hence:

Corollary 2.7. *Let $X \in \Gamma(S, \mathfrak{g}_\alpha)^\times$ and $u \in \Gamma(S, \mathcal{O}_S)^\times$. Then*

$$\alpha^*(u) = \exp((u^{-1} - 1)X^{-1}) \exp(X) \exp((u - 1)X^{-1}) \exp(-u^{-1}X).$$

Definition 2.8. *The morphism α^* is called the coroot associated with the root α .*

Remark 2.9. *If (G, T, α) is an S -elementary system, then so is $(G, T, -\alpha)$. Theorem 2.1 therefore gives a duality between $\mathfrak{g}_{-\alpha}$ and $\mathfrak{g}_\alpha$, and a coroot $(-\alpha)^*$. Taking the inverse of formula (F) gives immediately*

$$\langle X, Y \rangle = \langle Y, X \rangle, \quad (-\alpha)^* = -\alpha^*.$$

We now pass to the Lie algebra of G . The root α and the coroot α^* define linear forms

$$\mathcal{O}_S \xrightarrow{\bar{\alpha}^*} \mathfrak{t} \xrightarrow{\bar{\alpha}} \mathcal{O}_S.$$

Put

$$H_\alpha = \bar{\alpha}^*(1).$$

We call $\bar{\alpha}$ the infinitesimal root associated with α , and H_α the corresponding infinitesimal coroot.

Lemma 2.10. *Let $S' \rightarrow S$, and let*

$$X, X' \in W(\mathfrak{g}_\alpha)(S'), \quad H \in W(\mathfrak{t})(S'), \quad Y, Y' \in W(\mathfrak{g}_{-\alpha})(S'), \quad t \in T(S').$$

Then

- (1) $\text{Ad}(t)H = H, \quad \text{Ad}(t)X = \alpha(t)X, \quad \text{Ad}(t)Y = \alpha(t)^{-1}Y;$
- (2) $\text{Ad}(\exp X)H = H - \bar{\alpha}(H)X, \quad \text{Ad}(\exp X)X' = X',$
 $\text{Ad}(\exp X)Y = Y + \langle X, Y \rangle H_\alpha - \langle X, Y \rangle X;$
- (2') $\text{Ad}(\exp Y)H = H + \bar{\alpha}(H)Y, \quad \text{Ad}(\exp Y)Y' = Y',$
 $\text{Ad}(\exp Y)X = X + \langle X, Y \rangle H_{-\alpha} - \langle X, Y \rangle Y;$
- (3) $[H, X] = \bar{\alpha}(H)X, \quad [H, Y] = -\bar{\alpha}(H)Y,$
 $[X, Y] = \langle X, Y \rangle H_\alpha;$
- (4) $H_{-\alpha} = -H_\alpha,$
- (5) $\bar{\alpha}(H_\alpha) = 2.$

The proof of these formulas is either immediate or an immediate consequence of formula (F) of 2.1.

Corollary 2.11. *Suppose that H_α is nonzero on every fiber; by (5), this holds in particular if 2 is invertible on S . Then*

$$X_\alpha \in \Gamma(S, \mathfrak{g}_\alpha)^\times, \quad X_{-\alpha} \in \Gamma(S, \mathfrak{g}_{-\alpha})^\times$$

are paired if and only if

$$[X_\alpha, X_{-\alpha}] = H_\alpha.$$

2.12. Let (G, T, α) be an S -elementary system. By 1.19,

$$\text{Centr}(G) = \ker(\alpha)$$

is a group of multiplicative type and finite type. If Q is a subgroup of multiplicative type of $\text{Centr}(G)$, then G/Q is affine over S (Exposé IX, 2.5), smooth over S (Exposé VI_B, 9.2), and has connected reductive fibers of semisimple rank 1 (Exposé XIX, 1.8).

Put $G' = G/Q$. It is a reductive S -group of semisimple rank 1, and $T' = T/Q$ is a maximal torus of G' . The open subset

$$U_{-\alpha} \cdot T \cdot U_\alpha$$

of G is stable under Q , and its quotient is isomorphic to

$$U_{-\alpha} \times_S (T/Q) \times_S U_\alpha.$$

If α' denotes the character of T' induced by α , the canonical morphism $G \rightarrow G'$ induces isomorphisms

$$\mathfrak{g}_\alpha \xrightarrow{\sim} \mathfrak{g}'_{\alpha'}, \quad \mathfrak{g}_{-\alpha} \xrightarrow{\sim} \mathfrak{g}'_{-\alpha'}.$$

In particular, α' is a root of G' relative to T' . Writing α/Q for the character

$$T/Q \longrightarrow \mathbb{G}_{m,S}$$

induced by α , we obtain:

Lemma 2.13. *If Q is a subgroup of multiplicative type of $\ker(\alpha)$, then*

$$(G/Q, T/Q, \alpha/Q)$$

is an elementary system.

Lemma 2.14. *Under the preceding conditions, the following diagrams are commutative:*

$$\begin{array}{ccccc}
 W(\mathfrak{g}^\alpha) & \xrightarrow{\text{exp}} & G & \xleftarrow{\text{exp}} & W(\mathfrak{g}^{-\alpha}) \\
 \text{can} \downarrow \sim & & \text{can} \downarrow & & \text{can} \downarrow \sim \\
 W(\mathfrak{g}'^{\alpha'}) & \xrightarrow{\text{exp}} & G' & \xleftarrow{\text{exp}} & W(\mathfrak{g}'^{-\alpha'}) \\
 & & & & \\
 \mathfrak{g}^\alpha \otimes \mathfrak{g}^{-\alpha} & \xrightarrow{\sim} & \mathcal{O}_S & & \\
 \text{can} \downarrow \sim & & \downarrow \text{id} & & \\
 \mathfrak{g}'^{\alpha'} \otimes \mathfrak{g}'^{-\alpha'} & \xrightarrow{\sim} & \mathcal{O}_S & & \\
 & & & & \\
 & & T & & \\
 \mathbb{G}_{m,S} \nearrow \alpha^* & & \downarrow \text{can} & & \searrow \alpha \\
 & & T' & & \\
 \mathbb{G}_{m,S} \searrow \alpha'^* & & \nearrow \alpha' & &
 \end{array}$$

3 The Weyl group

Notations 3.0. ²³ If (G, T, α) is an S -elementary system, write

$$N = \text{Norm}_G(T), \quad W = \text{Norm}_G(T)/T$$

(cf. Exposé XIX, 6.3). The group N is a closed subgroup of G , smooth over S . Write

$$N^\times = N - T$$

for the open subscheme of N induced on the complement of T .²⁴

Let R be the unique maximal torus of $\text{Ker}(\alpha)$, and let T' be the image of

$$\alpha^* : \mathbb{G}_{m,S} \longrightarrow T;$$

it is a one-dimensional subtorus of T . The morphism

$$T' \times_S R \longrightarrow T$$

induced by multiplication in T is surjective, hence faithfully flat. Indeed, it is enough to check this on the geometric fibers, where it follows at once from the formula

$$\alpha \circ \alpha^* = 2.$$

²³The editors added the number 3.0 for later references.

²⁴The editors replaced Q by the more suggestive notation $N^\times$.

Theorem 3.1. *With the preceding notation:*

- (i) W is isomorphic to the constant group $(\mathbb{Z}/2\mathbb{Z})_S$.
- (ii) $N^\times$ is a locally trivial principal homogeneous bundle under T , on the left by the law $(t, q) \mapsto tq$, and on the right by the law $(q, t) \mapsto qt$.
- (iii) One has

$$\text{int}(w)t = t \cdot \alpha^*(\alpha(t)^{-1})$$

for $w \in N^\times(S')$, $t \in T(S')$, and $S' \rightarrow S$. In the decomposition $T_{S'} = T'_{S'} \cdot R_{S'}$, $\text{int}(w)$ induces the identity on $R_{S'}$ and inversion on $T'_{S'}$. One has the relations

$$\alpha \circ \text{int}(w) = \alpha^{-1}, \quad \text{int}(w) \circ \alpha^* = (\alpha^*)^{-1}.$$

- (iv) For $X \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \rightarrow S$, set

$$w_\alpha(X) = \exp(X) \exp(-X^{-1}) \exp(X).$$

Then $w_\alpha(X) \in N^\times(S')$, and the morphism $w_\alpha: W(\mathfrak{g}^\alpha)^\times \rightarrow N^\times$ so defined satisfies

$$w_\alpha(zX) = \alpha^*(z)w_\alpha(X) = w_\alpha(X)\alpha^*(z)^{-1}$$

for $z \in \mathbb{G}_m(S')$, $X \in W(\mathfrak{g}^\alpha)^\times(S')$, and $S' \rightarrow S$.

- (v) For $X, Y \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \rightarrow S$, one has, with the notation of 2.6,

$$w_\alpha(X)w_\alpha(Y) = \alpha^\vee(-XY^{-1}).$$

²⁵ In particular,

$$\begin{aligned} w_\alpha(X)^2 &= \alpha^*(-1) \in {}_2T(S) \cap \text{Centr}(G)(S), \\ w_\alpha(X)^{-1} &= w_\alpha(-X) = \alpha^*(-1)w_\alpha(X). \end{aligned}$$

- (vi) If, similarly, one defines for $Y \in W(\mathfrak{g}^{-\alpha})^\times(S')$

$$w_{-\alpha}(Y) = \exp(Y) \exp(-Y^{-1}) \exp(Y),$$

then, in addition to the formulas analogous to the preceding ones,

$$w_{-\alpha}(X^{-1}) = w_\alpha(X)^{-1} = w_\alpha(-X), \quad w_\alpha(X)w_{-\alpha}(Y) = \alpha^*(XY).$$

Proof. Assertion (i) was already proved in Exposé XIX, 2.4. It follows at once that $N^\times$ is indeed a principal homogeneous bundle under T for the laws defined in (ii); its local triviality for the Zariski topology follows in particular from (iv).

Let us prove (iii). If $w \in N^\times(S)$, it is clear that $\alpha \circ \text{int}(w)$ is a root of G relative to T , and hence is locally equal to α or $-\alpha$. Since it is $-\alpha$ on every fiber (Bible, 12-05, proof of the corollary to Proposition 1), one has

$$\alpha \circ \text{int}(w) = -\alpha.$$

By transport of structure, it follows that

$$-\alpha^* = \text{int}(w)^{-1} \circ \alpha^* = \text{int}(w) \circ \alpha^*,$$

²⁵The editors corrected the first formula of the original.

since $\text{int}(w)^2 = \text{int}(w^2)$ and w^2 is a section of T . Thus $\text{int}(w)$ induces inversion on T' ; since R is central, it induces the identity on R . The formula in (iii) defines a morphism $T \rightarrow T$ having the same properties, and therefore coincides with $\text{int}(w)$.

Let us prove (iv). Successively,

$$\begin{aligned} w_\alpha(X)tw_\alpha(X)^{-1} &= \exp(X)\exp(-X^{-1})\exp(X)t\exp(-X)\exp(X^{-1})\exp(-X) \\ &= \exp(X)\exp(-X^{-1})\exp(X - \alpha(t)X)\exp(\alpha(t)^{-1}X^{-1})\exp(-\alpha(t)X)t. \end{aligned}$$

Applying formula (F) gives

$$\begin{aligned} \exp(-X^{-1})\exp((1 - \alpha(t))X) &= \exp((\alpha(t)^{-1} - 1)X)\alpha^*(\alpha(t)^{-1}) \\ &\quad \cdot \exp(-\alpha(t)^{-1}X^{-1}). \end{aligned}$$

Substitution in the preceding relation yields

$$\begin{aligned} \text{int}(w_\alpha(X))t &= \exp(\alpha(t)^{-1}X)\alpha^*(\alpha(t)^{-1})\exp(-\alpha(t)X)t \\ &= \exp(aX)\alpha^*(\alpha(t)^{-1})t, \end{aligned}$$

where

$$a = \alpha(t)^{-1} - (\alpha \circ \alpha^*)(\alpha(t)^{-1})\alpha(t).$$

But $\alpha \circ \alpha^* = 2$, which gives $a = 0$ at once and shows that $w_\alpha(X) \in N^\times(S')$.

We now prove the second assertion of (iv). One has²⁶

$$\begin{aligned} \alpha^*(z)w_\alpha(X) &= \exp(z^2X)\exp(-z^{-2}X^{-1})\exp(z^2X)\alpha^*(z) \\ &= \exp(zX)\exp((z^2 - z)X)\exp(-z^{-2}X^{-1})\exp(z^2X)\alpha^*(z) \\ &= \exp(zX)\exp(-z^{-1}X^{-1})\alpha^*(z)^{-1}\exp((z^3 - z^2)X)\exp(z^2X)\alpha^*(z) \\ &= \exp(zX)\exp(-z^{-1}X^{-1})\exp(zX) = w_\alpha(zX). \end{aligned}$$

Let us prove (v). By the preceding result, the first formula in (v) follows at once from the second; we prove the latter:

$$\begin{aligned} w_\alpha(X)^2 &= \exp(X)\exp(-X^{-1})\exp(2X)\exp(-X^{-1})\exp(X) \\ &= \exp(X)\exp(-X^{-1})\exp(X^{-1})\alpha^*(-1)\exp(-2X)\exp(X) \\ &= \exp(X)\alpha^*(-1)\exp(-X) = \alpha^*(-1), \end{aligned}$$

since

$$\alpha(\alpha^*(-1)) = (-1)^2 = 1.$$

This also proves that $\alpha^*(-1) \in \text{Centr}(G)(S)$.

Finally, let us prove (vi). The first assertion is a special case of the second, so we prove the latter. Its two sides define morphisms

$$W(\mathfrak{g}^\alpha)^\times \times_S W(\mathfrak{g}^{-\alpha})^\times \longrightarrow G.$$

To prove that they coincide, it suffices to do so on a nonempty open subset of every fiber (Exposé XVIII, 1.4). We may therefore suppose that $1 + XY$ is invertible. Then,

²⁶The first equality follows from 1.5 (i), which, together with $\alpha \circ \alpha^* = 2$, gives $\alpha^*(z)\exp(X)\alpha^*(z)^{-1} = \exp(z^2X)$ and $\alpha^*(z)\exp(X^{-1})\alpha^*(z)^{-1} = \exp(z^{-2}X)$. The third equality follows from formula (F), and the fourth again from these formulas. Finally, an analogous calculation gives $w_\alpha(X)\alpha^*(z^{-1}) = w_\alpha(zX)$.

successively,

$$\begin{aligned}
w_\alpha(X)w_{-\alpha}(Y) &= \exp(X) \exp(-X^{-1}) \exp(X) \exp(Y) \\
&\quad \exp(-Y^{-1}) \exp(Y) \\
&= \exp(X) \exp(-X^{-1}) \exp\left(\frac{Y}{1+XY}\right) \alpha^*(1+XY) \\
&\quad \exp\left(\frac{X}{1+XY}\right) \exp(-Y^{-1}) \exp(Y) \\
&= \exp(X) \exp\left(\frac{-X^{-1}}{1+XY}\right) \alpha^*(1+XY) \exp\left(\frac{-Y^{-1}}{1+XY}\right) \exp(Y) \\
&= \exp(-X^{-2}Y^{-1}) \alpha^*\left(\frac{XY}{1+XY}\right) \exp(X+Y^{-1}) \alpha^*(1+XY) \\
&\quad \cdot \exp\left(\frac{-Y^{-1}}{1+XY}\right) \exp(Y) \\
&= \exp(-X^{-2}Y^{-1}) \alpha^*(XY) \exp\left(\frac{Y^{-1}+X}{(1+XY)^2}\right) \\
&\quad \exp\left(\frac{-Y^{-1}}{1+XY}\right) \exp(Y) \\
&= \alpha^*(XY) \exp(-Y) \exp(Y) = \alpha^*(XY).
\end{aligned}$$

Corollary 3.2. *Let $n \in \mathbb{Z}$, $n \neq 0$. For every $w \in G(S)$, the following conditions are equivalent:*

- (i) $w \in N^\times(S)$;
- (ii) $\text{int}(w) \circ n\alpha^* = -n\alpha^*$, where $(n\alpha^*)(z) = \alpha^*(z)^n$.

One has (i) $\Rightarrow$ (ii) by Theorem 3.1 (iii). Conversely, one may suppose that $N^\times$ has a section, and one is reduced to proving:

Lemma 3.3. *For $n \neq 0$, one has*

$$\text{Centr}_G(n\alpha^*) = T.$$

Indeed, the image T' of $n\alpha^*$ is a subtorus of G . It follows from Exposé XIX, 2.8 that $\text{Centr}_G(n\alpha^*)$ is a reductive subgroup of G containing T . On every fiber,

$$\text{Centr}_{G_s}(n\alpha_s^*) \neq G_s,$$

and hence

$$\text{Centr}_{G_s}(n\alpha_s^*) = T_s$$

by Exposé XIX, 1.6.3.²⁷ Therefore

$$\text{Centr}_G(n\alpha^*) = T,$$

since these are smooth subgroups of G .

²⁷The hypothesis $\text{Centr}_{G_s}(n\alpha_s^*) \neq G_s$ implies

$$\dim \text{Centr}_{G_s}(n\alpha_s^*) - \dim T_s < 2;$$

this difference is even by the cited result.

Remark 3.4. The construction of w_α , and the fact that $w_\alpha(X)$ normalizes T , depend only on formula (F). In particular, if G is an S -group satisfying the conditions of 2.2, then $\text{Norm}_G(T)$ differs from T on every fiber. Consequently, if G is an affine S -group with connected fibers satisfying the conditions of 2.2, then G is reductive of semisimple rank 1. Indeed, it is smooth in a neighborhood of the identity section, hence smooth, and the criterion of Exposé XIX, 1.11 applies.

3.5. Before stating the following theorem, let us make a few remarks. As usual, we identify $\mathfrak{g}^{-\alpha}$ with $(\mathfrak{g}^\alpha)^{\otimes -1}$. Similarly, we identify

$$\text{Hom}_{\mathcal{O}_S}(\mathfrak{g}^{-\alpha}, \mathfrak{g}^\alpha) \quad \text{with} \quad (\mathfrak{g}^\alpha)^{\otimes 2},$$

and hence

$$\text{Isom}_{\mathcal{O}_S\text{-mod.}}(W(\mathfrak{g}^{-\alpha}), W(\mathfrak{g}^\alpha)) \simeq W((\mathfrak{g}^\alpha)^{\otimes 2})^\times.$$

If $w \in N^\times(S)$, then $\text{Ad}(w)$ permutes $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ (3.1 (iii)), and therefore defines an isomorphism

$$a_\alpha(w): \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathfrak{g}^\alpha.$$

We thus identify it with a section

$$a_\alpha(w) \in \Gamma(S, (\mathfrak{g}^\alpha)^{\otimes 2})^\times.$$

This construction is compatible with base change and therefore defines a morphism

$$a_\alpha: N^\times \longrightarrow W((\mathfrak{g}^\alpha)^{\otimes 2})^\times$$

such that

$$a_\alpha(w)Y = \text{Ad}(w)Y$$

for every $w \in N^\times(S')$, $Y \in \Gamma(S', \mathfrak{g}^{-\alpha})^\times$, and $S' \rightarrow S$.

Theorem 3.6. (i) For every $S' \rightarrow S$, $w \in N^\times(S')$, and $Y \in W(\mathfrak{g}^{-\alpha})(S')$, one has

$$\text{int}(w) \exp(Y) = \exp(a_\alpha(w)Y).$$

(ii) One has

$$a_\alpha(tw) = \alpha(t)a_\alpha(w), \quad a_\alpha(wt) = \alpha(t)^{-1}a_\alpha(w).$$

(iii) If one similarly defines

$$a_{-\alpha}: N^\times \longrightarrow W((\mathfrak{g}^{-\alpha})^{\otimes 2})^\times,$$

then

$$a_{-\alpha}(w) = a_\alpha(w)^{-1}.$$

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(iv) For every $X \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \rightarrow S$, one has

$$a_\alpha(w_\alpha(X)) = -X^2.$$

²⁸That is, $a_{-\alpha}(w)$ and $a_\alpha(w)$ are paired; cf. 2.6.1.

Proof. Assertion (i) is immediate from the characterization of the morphisms $\exp$ given in 1.5. Assertion (ii) is immediate, as is (iii). Let us prove (iv). Let

$$X \in \Gamma(S', \mathfrak{g}^\alpha)^\times, \quad Z \in \Gamma(S', \mathfrak{g}^\alpha).$$

By definition,²⁹

$$\begin{aligned} a_\alpha(w_\alpha(X))^{-1}(Z) &= \text{Ad}(w_\alpha(X))(Z) \\ &= \text{Ad}(\exp(X)) \text{Ad}(\exp(-X^{-1})) \text{Ad}(\exp(X))(Z). \end{aligned}$$

Applying formulas (2') and (2) of Lemma 2.10, together with

$$H_{-\alpha} = -H_\alpha, \quad \bar{\alpha}(H_\alpha) = 2$$

(loc. cit., (4) and (5)), and $\langle X, X^{-1} \rangle = 1$ (2.6), the expression on the right is, successively,

$$\begin{aligned} &\text{Ad}(\exp(X)) \text{Ad}(\exp(-X^{-1}))(Z) \\ &= \text{Ad}(\exp(X))(Z + \langle X^{-1}, Z \rangle (H_\alpha - X^{-1})) \\ &= Z + \langle X^{-1}, Z \rangle (H_\alpha - 2X - X^{-1} - H_\alpha + X) \\ &= Z - \langle X^{-1}, Z \rangle X - \langle X^{-1}, Z \rangle X^{-1}. \end{aligned}$$

But

$$Z = \langle X^{-1}, Z \rangle X, \quad \langle X^{-1}, Z \rangle X^{-1} = X^{-2}Z.$$

Thus

$$a_\alpha(w_\alpha(X))^{-1} = -X^{-2},$$

and consequently

$$a_\alpha(w_\alpha(X)) = -X^2.$$

Corollary 3.7. *In particular, one has*

$$\text{int}(w_\alpha(X)) \exp(X) = \exp(-X^{-1}).$$

Hence, by the definition of $w_\alpha(X)$,

$$w_\alpha(X) \exp(X) w_\alpha(X)^{-1} = \exp(-X) w_\alpha(X) \exp(-X),$$

and therefore, by an immediate calculation,

$$(w_\alpha(X) \exp(X))^3 = e.$$

Corollary 3.8. *Let $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, and let $n \in \mathbb{Z}$, $n \neq 0$. Then $w_\alpha(X)$ is the unique section $w \in G(S)$ satisfying*

$$(i) \text{ int}(w) \circ n\alpha^* = -n\alpha^*;$$

$$(ii) (w \exp(X))^3 = e.$$

We know that $w_\alpha(X)$ satisfies these conditions. Conversely, let $w \in G(S)$ satisfy (i) and (ii). By 3.2 and 3.1 (ii), there exists $t \in T(S)$ such that

$$w = w_\alpha(X)t.$$

²⁹The editors supplied the details that follow.

Set $u = \exp(X)$. Then

$$\begin{aligned} wuw^{-1} &= w_\alpha(X)t \exp(X)t^{-1}w_\alpha(X)^{-1} \\ &= \exp(-\alpha(t)X^{-1}), \end{aligned}$$

whereas

$$\begin{aligned} u^{-1}wu^{-1} &= \exp(-X)w_\alpha(X)t \exp(-X) \\ &= \exp(-X)w_\alpha(X) \exp(-X) \exp(X - \alpha(t)X)t \\ &= \exp(-X^{-1}) \exp(X - \alpha(t)X)t \\ &= \exp(-X^{-1})t \exp(H). \end{aligned}$$

Now

$$(wu)^3 = e \iff wuw^{-1} = u^{-1}wu^{-1}.$$

Comparing the two decompositions of this element in

$$U_{-\alpha} \cdot T \cdot U_\alpha$$

gives $t = e$.

Remark 3.9. We may summarize a number of the results of this section by the following diagram of principal homogeneous bundles (on the left):

$$\begin{array}{ccccc} W(\mathfrak{g}^\alpha)^\times & \xrightarrow{w_\alpha} & N^\times & \xrightarrow{a_\alpha} & W((\mathfrak{g}^\alpha)^{\otimes 2})^\times \\ \mathbb{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbb{G}_{m,S} \end{array}$$

Notice that a_α is faithfully flat (since α is), and that w_α is a monomorphism if and only if α^* is a monomorphism. We leave to the reader the task of writing down the corresponding diagrams for right principal-bundle structures, analogous diagrams for the root $-\alpha$, and of studying the relations among these various diagrams.

Lemma 3.10. *Let S be a scheme, and let $q > 0$ be an integer such that $x \mapsto x^q$ defines an endomorphism of $\mathbb{G}_{a,S}$. Let (G, T, α) and (G', T', α') be two elementary S -systems, and let $f: G \rightarrow G'$ be a morphism of S -groups. Let*

$$h: (\mathfrak{g}^\alpha)^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}$$

be an isomorphism of $\mathcal{O}_S$ -modules, and let

$$h^\vee: (\mathfrak{g}^{-\alpha})^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{-\alpha'}$$

be the contragredient isomorphism. Suppose that, for every $S' \rightarrow S$ and every $X \in W(\mathfrak{g}^\alpha)(S')$,

$$f(\exp(X)) = \exp(h(X^q)).$$

Then the following conditions are equivalent:

- (i) $f(\alpha^*(z)) = \alpha'^*(z)^q$.
- (ii) $f(w_\alpha(Z)) = w_{\alpha'}(h(Z^q))$.
- (iii) $f(\exp(Y)) = \exp(h^\vee(Y^q))$.

Each condition is to be read as holding for every $S' \rightarrow S$ and every

$$z \in \mathbb{G}_m(S'), \quad Z \in W(\mathfrak{g}^\alpha)^\times(S'), \quad Y \in W(\mathfrak{g}^{-\alpha})(S'),$$

respectively. Indeed, (i) implies (ii) by 3.8, (ii) implies (iii) by 3.7, and (iii) implies (i) by 2.7.

Proposition 3.11. *Let S be a scheme, and let $q \in \mathbb{Z}$, $q > 0$, be such that $x \mapsto x^q$ defines an endomorphism of $\mathbb{G}_{a,S}$. Let (G, T, α) and (G', T', α') be two elementary S -systems, and let $f: G \rightarrow G'$ be a morphism of S -groups. The following conditions on f are equivalent:*

(i) *The restriction of f to T factors through a morphism $f_T: T \rightarrow T'$ making the diagram*

$$\begin{array}{ccccc} \mathbb{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbb{G}_{m,S} \\ q \downarrow & & f_T \downarrow & & q \downarrow \\ \mathbb{G}_{m,S} & \xrightarrow{\alpha'^*} & T' & \xrightarrow{\alpha'} & \mathbb{G}_{m,S} \end{array}$$

commutative.

(ii) *There exists a unique isomorphism of $\mathcal{O}_S$ -modules*

$$h: (\mathfrak{g}^\alpha)^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}$$

such that

$$f(\exp(X)) = \exp(h(X^q)), \quad f(\exp(Y)) = \exp(h^\vee(Y^q))$$

for every $S' \rightarrow S$, $X \in W(\mathfrak{g}^\alpha)(S')$, and $Y \in W(\mathfrak{g}^{-\alpha})(S')$. Consequently, f also satisfies the equivalent conditions of 3.10.

We first prove that (ii) implies (i). By 3.10, condition (ii) gives

$$f \circ \alpha^* = q\alpha'^*;$$

hence, by 3.3, $f|_T$ factors through T' . It remains to prove

$$\alpha'(f(t)) = \alpha(t)^q,$$

which follows at once from the fact that f induces a morphism of groups

$$T \cdot U_\alpha \longrightarrow T' \cdot U_{\alpha'}.$$

We now prove that (i) implies (ii). Let

$$X \in \Gamma(S, \mathfrak{g}^\alpha), \quad Y \in \Gamma(S, \mathfrak{g}^{-\alpha}),$$

and set

$$p_+(x) = f(\exp(xX)), \quad p_-(y) = f(\exp(yY)).$$

These are morphisms of groups

$$p_+, p_-: \mathbb{G}_{a,S} \longrightarrow G.$$

Now

$$\begin{aligned} \text{int}(\alpha'^*(z))^q(p_+(x)) &= \text{int}(f_T(\alpha^*(z)))(f(\exp(xX))) \\ &= f(\text{int}(\alpha^*(z))(\exp(xX))) \\ &= f(\exp(z^2xX)) = p_+(z^2x). \end{aligned}$$

Applying Lemma 1.2 with $Q = \alpha'^*(\mathbb{G}_{m,S})$, we obtain a section

$$X' \in \Gamma(S, \mathfrak{g}'^{\alpha'})$$

such that

$$f(\exp(xX)) = p_+(x) = \exp(x^q X').$$

Similarly, there exists a section

$$Y' \in \Gamma(S, \mathfrak{g}'^{-\alpha'})$$

such that

$$f(\exp(yY)) = \exp(y^q Y').$$

Since f is a group morphism, it respects formula (F); hence

$$X^q Y^q = (XY)^q = X' Y'.$$

It follows readily that $X^q \mapsto X'$ and $Y^q \mapsto Y'$ define the announced isomorphisms h and $h^\vee$.

Proposition 3.12. *Let (G, T, α) be an elementary S -system and let $w \in N^\times(S)$. Put*

$$\Omega_0 = \Omega \cap \text{int}(w^{-1})(\Omega).$$

Let d be the function on Ω defined by

$$d(\exp(Y) \cdot t \cdot \exp(X)) = \alpha(t)^{-1} + XY.$$

Then $\Omega_0 = \Omega_d$, and, for $\exp(Y) \cdot t \cdot \exp(X) \in \Omega_0(S')$, one has the following formula, where $z = d(\exp(Y) \cdot t \cdot \exp(X))$:

$$(\star) \quad \text{int}(w)(\exp(Y) \cdot t \cdot \exp(X)) = \exp(z^{-1} a_\alpha(w)^{-1} X) \cdot t \alpha^*(z) \cdot \exp(z^{-1} a_\alpha(w) Y).$$

Moreover,

$$d \circ \text{int}(w) = d^{-1}.$$

Indeed, one immediately has³⁰

$$\begin{aligned} \text{int}(w)(\exp(Y) \cdot t \cdot \exp(X)) &= \exp(a_\alpha(w) Y) \cdot t \alpha^*(\alpha(t)^{-1}) \cdot \exp(a_\alpha(w)^{-1} X) \\ &= \exp(a_\alpha(w) Y) \cdot \exp(\alpha(t) a_\alpha(w)^{-1} X) \cdot t \alpha^*(\alpha(t)^{-1}). \end{aligned}$$

By 2.1, this is a section of Ω if and only if $1 + \alpha(t)XY$ is invertible, which proves $\Omega_0 = \Omega_d$. Applying formula (F) of the same paragraph then gives $(\star)$ by an immediate computation. Finally, $(\star)$ gives

$$\begin{aligned} (d \circ \text{int}(w))(\exp(Y) \cdot t \cdot \exp(X)) &= \alpha(t \alpha^*(z))^{-1} + z^{-2} XY \\ &= z^{-2} (\alpha(t)^{-1} + XY) = z^{-1}, \end{aligned}$$

which proves the last assertion.

N.B. The function d is independent of the choice of w .

³⁰The editors corrected the original by interchanging $a_\alpha(w)$ and $a_\alpha(w)^{-1}$, and supplied the details of the proof of $d \circ \text{int}(w) = d^{-1}$.

4 The Isomorphism Theorem

Theorem 4.1. *Let S be a scheme and let $q \in \mathbb{Z}$, $q > 0$, be such that $x \mapsto x^q$ is an endomorphism of $\mathbb{G}_{a,S}$. Let (G, T, α) and (G', T', α') be two elementary S -systems. Let*

$$h: (\mathfrak{g}^\alpha)^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}, \quad h^\vee: (\mathfrak{g}^{-\alpha})^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{-\alpha'}$$

be mutually contragredient isomorphisms. Let $f_T: T \rightarrow T'$ be a morphism of S -groups making the following diagram commute:

$$\begin{array}{ccc} \mathbb{G}_{m,S} & \xrightarrow{q} & \mathbb{G}_{m,S} \\ \alpha^* \downarrow & & \downarrow \alpha'^* \\ T & \xrightarrow{f_T} & T' \\ \alpha \downarrow & & \downarrow \alpha' \\ \mathbb{G}_{m,S} & \xrightarrow{q} & \mathbb{G}_{m,S} \end{array}$$

There exists a unique morphism of S -groups $f: G \rightarrow G'$ extending f_T and satisfying

$$f(\exp(X)) = \exp(h(X^q))$$

for every $S' \rightarrow S$ and $X \in W(\mathfrak{g}^\alpha)(S')$. Moreover, this morphism also satisfies

$$f(\exp(Y)) = \exp(h^\vee(Y^q)), \quad f(w_\alpha(Z)) = w_{\alpha'}(h(Z^q))$$

for every $S' \rightarrow S$, $Y \in \Gamma(S', \mathfrak{g}^{-\alpha})$, and $Z \in \Gamma(S', \mathfrak{g}^\alpha)^\times$.

If $f: G \rightarrow G'$ extends f_T , then

$$f \circ \alpha^* = (\alpha'^*)^q.$$

If, in addition, f satisfies the second displayed condition in the statement, then it satisfies the other two conditions as well by 3.10. It follows that f is determined on Ω by

$$f(\exp(Y)t \exp(X)) = \exp(h^\vee(Y^q)) f_T(t) \exp(h(X^q)).$$

Since Ω is schematically dense in G , this already proves the uniqueness of f . To prove existence, it is enough, by Exposé XVIII, 2.3, to show that the preceding formula defines a “generically multiplicative” morphism from Ω to G' . By 2.4, this amounts to checking

$$\alpha' \circ f = \alpha^q,$$

which follows from the fact that f extends f_T .

Scholium 4.2. Theorem 4.1 may also be interpreted as follows. Let $\mathcal{E}$ be the category of elementary S -systems, and let $\mathcal{D}$ be the category of tuples

$$(\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S}, L),$$

where T is a torus, α and α^* are group morphisms such that $\alpha \circ \alpha^* = 2$, and L is an invertible $\mathcal{O}_S$ -module. The reader may specify the morphisms in the two categories under consideration. Define a functor

$$\begin{aligned} \mathcal{E} &\longrightarrow \mathcal{D}, \\ (G, T, \alpha) &\longmapsto (\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S}, \mathfrak{g}^\alpha). \end{aligned}$$

The preceding theorem says that this functor is fully faithful. In fact, it is an equivalence of categories, as will be seen in the next section. We already have:

Corollary 4.3. *If $q = 1$ and f_T is an isomorphism, then f is an isomorphism.*

Corollary 4.4. *If $q = 1$ and f_T is faithfully flat with kernel Q (cf. Exposé IX, 2.7), then f is faithfully flat and quasi-compact with kernel Q , and hence identifies G' with G/Q .*

Indeed, if f_T is faithfully flat with kernel Q , then

$$Q = \text{Ker}(f_T) \subset \text{Ker}(\alpha' \circ f_T) = \text{Ker}(\alpha).$$

Introduce the elementary S -system $(G/Q, T/Q, \alpha/Q)$ of 2.13. By 2.14, we are reduced to proving that f/Q induces an isomorphism from G/Q onto G' , which follows immediately from 4.3.

5 Examples of Elementary Systems; Applications

5.1. Let S be a scheme and L an invertible $\mathcal{O}_S$ -module. Consider the group GL_L over S defined by

$$\text{GL}_L(S') = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid \begin{array}{l} a, d \in \mathbb{G}_a(S'), \quad b \in W(L)(S'), \\ c \in W(L^{-1})(S'), \quad ad - bc \in \mathbb{G}_m(S') \end{array} \right\},$$

equipped with the usual matrix-multiplication law. It is locally isomorphic to $\text{GL}_{2,S}$. It is therefore an affine smooth S -group scheme with connected fibers.

Remark. Let L' and L'' be invertible sheaves on S such that

$$L = L' \otimes L''^{-1}.$$

Then there is an isomorphism of S -groups

$$\text{GL}_L \xrightarrow{\sim} \text{GL}(L' \oplus L'')$$

defined as follows: if x (respectively y) is a section of L' (respectively L'') on an open subset V of S , then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

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5.2. Denote by SL_L the closed subgroup of GL_L defined by $ad - bc = 1$. It too is an affine smooth S -group scheme with connected fibers; under the preceding isomorphism it is isomorphic to $\text{SL}(L' \oplus L'')$.

Likewise, consider the morphism

$$\mathbb{G}_{m,S} \longrightarrow \text{GL}_L, \quad z \longmapsto \begin{pmatrix} z & 0 \\ 0 & z \end{pmatrix}.$$

It is a central monomorphism. Passage to the quotient gives a smooth affine S -group P_L with connected fibers (cf. Exposé VIII, 5.7). By passing to the quotient in the isomorphism of the preceding remark, one sees that P_L identifies with the automorphism group of the projective bundle $\mathbb{P}(L' \oplus L'')$ (cf. EGA II, 4.2.7). Write i and p for the canonical morphisms

$$\text{SL}_L \xrightarrow{i} \text{GL}_L \xrightarrow{p} \text{P}_L.$$

Here i is a closed immersion, while p is faithfully flat and affine.

³¹The editors corrected $L'' \otimes L'^{-1}$ to $L' \otimes L''^{-1}$ and expanded the sentence that follows.

5.3. Consider the group morphisms

$$\begin{aligned} t_G: \mathbb{G}_{m,S}^2 &\longrightarrow \mathrm{GL}_L, & t_G(z, z') &= \begin{pmatrix} z & 0 \\ 0 & z' \end{pmatrix}, \\ t_S: \mathbb{G}_{m,S} &\longrightarrow \mathrm{SL}_L, & t_S(z) &= \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}, \\ t_P: \mathbb{G}_{m,S} &\longrightarrow \mathrm{P}_L, & t_P(z) &= p(t_G(z, 1)). \end{aligned}$$

They are group monomorphisms and define in each group a split torus of relative codimension 2. For $s \in S$, let

$$X \in \Gamma(s, L \otimes s)^\times.$$

The section $\begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix}$ of $\mathrm{GL}_{L,s}$ normalizes $t_G(\mathbb{G}_{m,s}^2)$ and does not centralize it. Exposé XIX, 1.6 therefore shows that GL_L is reductive of semisimple rank 1, with maximal torus $t_G(\mathbb{G}_{m,S}^2)$. The same argument shows that SL_L and P_L are reductive of semisimple rank 1, with respective maximal tori $t_S(\mathbb{G}_{m,S})$ and $t_P(\mathbb{G}_{m,S})$.

5.4. The Lie algebras of these groups, and the adjoint actions of the chosen maximal tori, are determined in the usual way. For GL_L , Exposé II, 4.8 immediately gives

$$\mathrm{Lie}(\mathrm{GL}_L/S) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, d \in \mathcal{O}_S, b \in L, c \in L^{-1} \right\},$$

with the usual bracket, and

$$\mathrm{Ad}(t_G(z, z')) \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & zz'^{-1}b \\ z'z^{-1}c & d \end{pmatrix}.$$

Put $\mathfrak{g} = \mathrm{Lie}(\mathrm{GL}_L/S)$, and define

$$\alpha_G: t_G(\mathbb{G}_{m,S}^2) \longrightarrow \mathbb{G}_{m,S}, \quad \alpha_G(t_G(z, z')) = zz'^{-1}.$$

The preceding identity shows that α_G is a root of GL_L relative to $t_G(\mathbb{G}_{m,S}^2)$, and that

$$u: L \longrightarrow \mathfrak{g}, \quad X \longmapsto \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \quad u^-: L^{-1} \longrightarrow \mathfrak{g}, \quad Y \longmapsto \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix}$$

identify L with $\mathfrak{g}^{\alpha_G}$ and L^{-1} with $\mathfrak{g}^{-\alpha_G}$, respectively. Thus

$$(\mathrm{GL}_L, t_G(\mathbb{G}_{m,S}^2), \alpha_G)$$

is an elementary S -system.

Similarly, setting

$$\alpha_S(t_S(z)) = z^2, \quad \alpha_P(t_P(z)) = z,$$

one finds elementary S -systems

$$(\mathrm{SL}_L, t_S(\mathbb{G}_{m,S}), \alpha_S), \quad (\mathrm{P}_L, t_P(\mathbb{G}_{m,S}), \alpha_P),$$

and corresponding identifications of L and L^{-1} with the root summands of their Lie algebras.

5.5. Set

$$\exp \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}.$$

This defines a morphism

$$W(\mathfrak{g}^{\alpha_G}) \longrightarrow \mathrm{GL}_L$$

whose tangent map is the canonical inclusion; it is therefore the unique morphism of this kind by 1.5. Likewise, set

$$\exp \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}.$$

Direct calculation of formula (F) gives

$$\left\langle \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix} \right\rangle = XY, \quad \alpha_G^*(z) = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} = t_G(z, z^{-1}).$$

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The open subset $N^\times = N_G^\times$ defined before 3.1 is

$$N_G^\times(S') = \left\{ \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix} \mid P \in W(\mathfrak{g}^\alpha)^\times(S'), Q \in W(\mathfrak{g}^{-\alpha})^\times(S') \right\}.$$

For $X \in W(\mathfrak{g}^\alpha)^\times(S')$,

$$w_{\alpha_G}(X) = \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix}.$$

Moreover, if

$$w = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix} \in N_G^\times(S'),$$

then

$$a_{\alpha_G}(w) = PQ^{-1} \in W((\mathfrak{g}^\alpha)^{\otimes 2})^\times(S');$$

that is, for $Y \in W(\mathfrak{g}^{-\alpha})^\times(S')$,

$$a_{\alpha_G}(w)(Y) = PQ^{-1}Y \in W(\mathfrak{g}^\alpha)^\times(S').$$

5.6. We leave the analogous calculations for SL_L and P_L to the reader. One obtains the same duality formula and the coroots

$$\alpha_S^*(z) = t_S(z), \quad \alpha_P^*(z) = t_P(z^2).$$

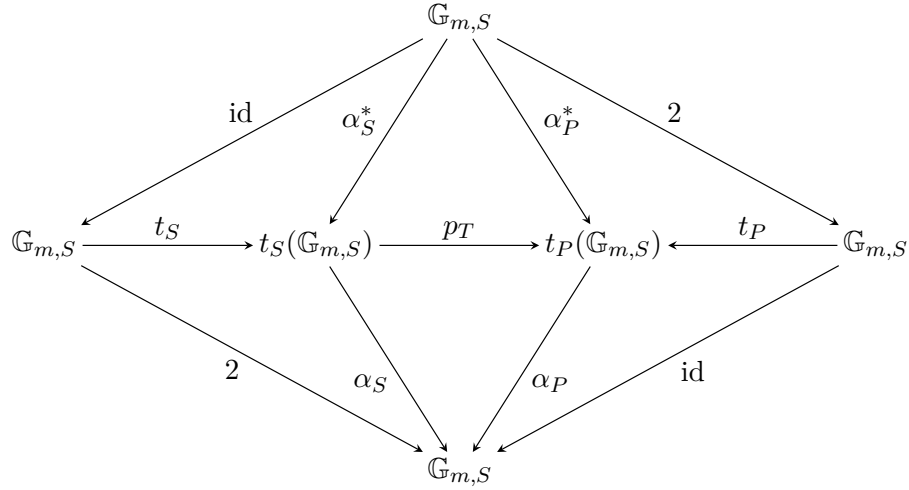
Let p_T be the morphism induced by $p: \mathrm{GL}_L \rightarrow \mathrm{P}_L$ on $t_S(\mathbb{G}_{m,S})$, namely

$$p_T(t_S(z)) = t_P(z^2).$$

Thus the following diagram is commutative: ³³

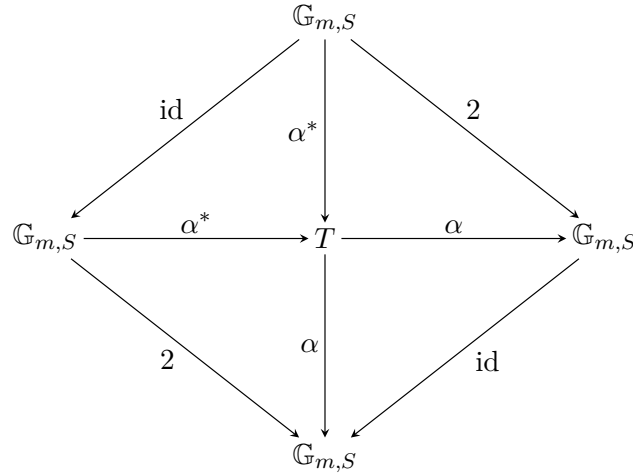
³²The editors expanded the calculations that follow.

³³Here t_S and t_P are isomorphisms.



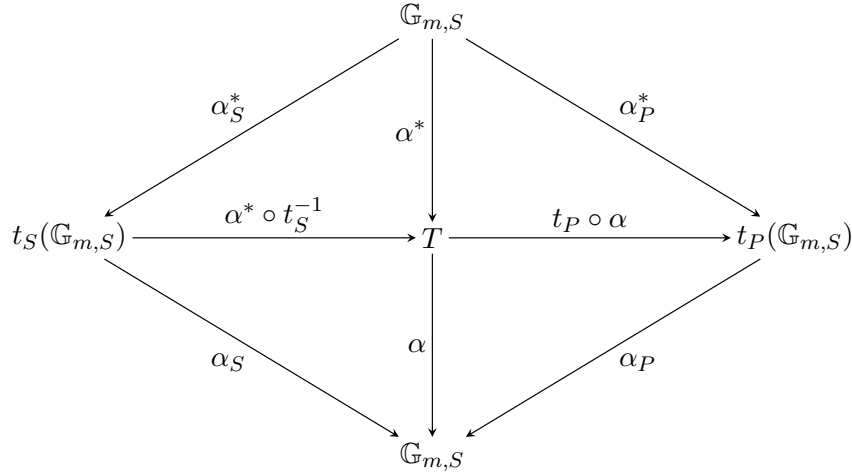
Its central portion is the commutative diagram of 4.1³⁴ associated with the canonical morphism $p \circ i: \text{SL}_L \rightarrow \text{P}_L$, which induces a morphism between the preceding elementary S -systems.

5.7. Let (G, T, α) now be an arbitrary elementary S -system. Consider the commutative diagram



Combining this diagram with that of 5.6 gives a commutative diagram

³⁴With $q = 1$.



Using 4.1, one obtains:

Proposition 5.8. *Let S be a scheme and let (G, T, α) be an elementary S -system. Put $L = \mathfrak{g}^\alpha$, so that $L^{-1} = \mathfrak{g}^{-\alpha}$.*

(i) *There exists a unique group morphism $f: \mathrm{SL}_L \rightarrow G$ satisfying the following equivalent conditions:*

$$\begin{aligned}
 \text{(a)} \quad & f \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} = \alpha^*(z), & f \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X); \\
 \text{(b)} \quad & f \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), & f \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix} = \exp(Y); \\
 \text{(c)} \quad & f \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), & f \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix} = w_\alpha(X).
 \end{aligned}$$

(ii) *There exists a unique group morphism $g: G \rightarrow \mathrm{P}_L$ satisfying*

$$g(t) = \begin{pmatrix} \alpha(t) & 0 \\ 0 & 1 \end{pmatrix}, \quad g(\exp(X)) = p \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}.$$

Moreover,

$$g(\exp(Y)) = p \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}, \quad g(w_\alpha(X)) = p \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix}.$$

The morphism g is faithfully flat and quasi-compact, with

$$\mathrm{Ker}(g) = \mathrm{Ker}(\alpha) = \mathrm{Centr}(G),$$

and $g \circ f$ is the canonical morphism $\mathrm{SL}_L \rightarrow \mathrm{P}_L$.

Notice that condition (b) in (i) gives an explicit description of the duality between $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$.

Corollary 5.9. *Let (G, T, α) be an elementary S -system. The subgroups*

$$T \cdot U_\alpha, \quad T \cdot U_{-\alpha}, \quad U_\alpha, \quad U_{-\alpha}$$

are closed.

Since U_α is a closed subgroup scheme of $T \cdot U_\alpha$, it suffices to prove the assertion for the latter. By Noether's theorem (Exposé IV, 5.3.1 and 6.4.1), it is enough to show that

$$(T \cdot U_\alpha) / \text{Ker}(\alpha)$$

is a closed subgroup of $G / \text{Ker}(\alpha)$. By 5.8, this reduces to the evident fact that the subgroup of P_L defined by $c = 0$ is closed; one may equivalently work in GL_L , by another application of Noether's theorem. Consequently, the exponential morphisms of Theorem 1.5(i) are closed immersions.

N.B. The corollary also follows from the fact that $T \cdot U_\alpha$ and $T \cdot U_{-\alpha}$ are “Borel subgroups” of G (cf. Exposé XII, 7.10).

5.10. Let L be an invertible $\mathcal{O}_S$ -module, and let

$$\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S}$$

be a diagram of groups, where T is a torus,³⁵ such that $\alpha \circ \alpha^* = 2$. Let R be the maximal torus of $\text{Ker}(\alpha)$, and put $K = \alpha^{*-1}(R)$. Then K is a subgroup of multiplicative type of $\mathbb{G}_{m,S}$; since $\alpha \circ \alpha^* = 2$, it is even a subgroup of $\mu_{2,S}$. In particular,

$$K \longrightarrow \text{SL}_L, \quad z \longmapsto \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}$$

is central. Hence there is a central group monomorphism

$$K \longrightarrow R \times \text{SL}_L, \quad z \longmapsto \left(\alpha^*(z), \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} \right).$$

Let

$$G = (R \times \text{SL}_L) / K.$$

This is an affine smooth group over S , with connected fibers. The sequence

$$1 \longrightarrow K \longrightarrow R \times t_S(\mathbb{G}_{m,S}) \xrightarrow{u} T \longrightarrow 1, \quad u(x, t_S(z)) = x\alpha^*(z),$$

is exact. The image of $R \times t_S(\mathbb{G}_{m,S})$ in G is therefore a torus T' isomorphic to T . If α' is the character of T' transported from α , one readily checks that (G, T', α') is an elementary S -system, that $\mathfrak{g}^{\alpha'}$ is isomorphic to L , and that α'^* is obtained from α^* under the isomorphism $T \xrightarrow{\sim} T'$. Thus the object

$$\left(\mathbb{G}_{m,S} \xrightarrow{\alpha'^*} T' \xrightarrow{\alpha'} \mathbb{G}_{m,S}, \mathfrak{g}^{\alpha'} \right)$$

of the category $\mathcal{D}$ defined in 4.2 is isomorphic to

$$\left(\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S}, L \right).$$

We have therefore proved:

Theorem 5.11. *With the notation of 4.2, the functor*

$$(G, T, \alpha) \longmapsto \left(\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S}, \mathfrak{g}^\alpha \right)$$

is an equivalence of categories between $\mathcal{E}$ and $\mathcal{D}$.

³⁵This qualification is supplied by the editors.

6 Generators and Relations for an Elementary System

6.1. Let S be a scheme and let (G, T, α) be an elementary S -system. Let

$$X \in W(\mathfrak{g}^\alpha)^\times(S), \quad u = \exp(X).$$

As observed in 3.8, the element $w = w_\alpha(X)$ satisfies, in particular,

$$(wu)^3 = e.$$

³⁶ Let s_α denote the automorphism of T induced by $\text{int}(w)$. By Theorem 3.1(iii), for every $S' \rightarrow S$ and every $t \in T(S')$,

$$s_\alpha(t) = \text{int}(w)(t) = t \alpha^*(\alpha(t)^{-1}).$$

Theorem 6.2. *Let H be a sheaf of groups over S for the fppf topology. Let*

$$f_T: T \longrightarrow H, \quad f_\alpha: U_\alpha \longrightarrow H$$

be group morphisms, and let $h \in H(S)$. There exists a (necessarily unique) group morphism

$$f: G \longrightarrow H$$

extending f_T and f_α and satisfying $f(w) = h$ if and only if the following conditions hold:

(i) *For every $S' \rightarrow S$, $t \in T(S')$, and $x \in U_\alpha(S')$,*

$$(1) \quad f_T(t)f_\alpha(x)f_T(t)^{-1} = f_\alpha(txt^{-1}) = f_\alpha(x^{\alpha(t)}).$$

Equivalently, f_T and f_α extend to a group morphism $T \cdot U_\alpha \rightarrow H$.

(ii) *For every $S' \rightarrow S$ and $t \in T(S')$,*

$$(2) \quad hf_T(t)h^{-1} = f_T(s_\alpha(t)) = f_T(t \alpha^*(\alpha(t)^{-1})).$$

(iii) *The following two relations hold in $H(S)$:*

$$(3) \quad h^2 = f_T(\alpha^*(-1)),$$

$$(4) \quad (hf_\alpha(u))^3 = e.$$

Proof. Write the laws on U_α and $U_{-\alpha}$ additively, while writing their vector-space structures multiplicatively. If f has the asserted properties, then for $y \in U_{-\alpha}(S')$ one necessarily has

$$f(y) = f(w^{-1}yw^{-1}w) = hf_\alpha(w^{-1}yw)h^{-1}.$$

Define a morphism $f_{-\alpha}: U_{-\alpha} \rightarrow H$ by

$$(*_1) \quad f_{-\alpha}(y) = hf_\alpha(w^{-1}yw)h^{-1}.$$

It is a group morphism. On the big cell Ω , the morphism f is forced to satisfy

$$f(ytx) = f_{-\alpha}(y)f_T(t)f_\alpha(x).$$

³⁶The editors added the sentence that follows.

This proves uniqueness. The conditions in the statement are manifestly necessary; let us prove that they are sufficient.

Relation (4) gives

$$hf_\alpha(u)h^{-1}h^2 = f_\alpha(-u)h^{-1}f_\alpha(-u).$$

By (3) and (1), $h^2 = f_T(\alpha^*(-1))$ commutes with $f_\alpha(-u)$, and hence

$$hf_\alpha(u)h^{-1} = f_\alpha(-u)hf_\alpha(-u).$$

By definition,

$$hf_\alpha(u)h^{-1} = f_{-\alpha}(wuw^{-1}),$$

whereas 3.7 gives

$$(*_2) \quad wuw^{-1} = -\tilde{u},$$

with $\tilde{u}$ the element paired with u . Therefore

$$(*_3) \quad f_{-\alpha}(-\tilde{u}) = f_\alpha(-u)hf_\alpha(-u).$$

Let t be a section of T over a variable $S' \rightarrow S$. Apply $\text{int}(f_T(t))$ to this identity. On the left one obtains³⁷

$$\begin{aligned} f_T(t)f_{-\alpha}(-\tilde{u})f_T(t)^{-1} &= f_T(t)hf_\alpha(u)h^{-1}f_T(t)^{-1} \\ &= h(h^{-1}f_T(t)h)f_\alpha(u)(h^{-1}f_T(t)^{-1}h)h^{-1} \\ &= hf_T(s_\alpha(t))f_\alpha(u)f_T(s_\alpha(t))^{-1}h^{-1} \\ &= hf_\alpha(\alpha(s_\alpha(t))u)h^{-1} \\ &= hf_\alpha(\alpha(t)^{-1}u)h^{-1} \\ &= f_{-\alpha}(-\alpha(t)^{-1}\tilde{u}). \end{aligned}$$

Here we used (2), (1), $s_\alpha(t) = t\alpha^*(\alpha(t)^{-1})$, and $\alpha \circ \alpha^* = 2$.

The right-hand side of $(*_3)$, after conjugation, becomes

$$f_\alpha(-\alpha(t)u)f_T(t)hf_T(t)^{-1}h^{-1}hf_\alpha(-\alpha(t)u).$$

Since

$$hf_T(t)^{-1}h^{-1} = f_T(s_\alpha(t^{-1})) = f_T(t\alpha^*(\alpha(t))),$$

this equals

$$f_\alpha(-\alpha(t)u)f_T(\alpha^*(\alpha(t)))hf_\alpha(-\alpha(t)u).$$

Comparison gives

$$f_{-\alpha}(-\alpha(t)^{-1}\tilde{u}) = f_\alpha(-\alpha(t)u)f_T(\alpha^*(\alpha(t)))hf_\alpha(-\alpha(t)u).$$

Since $\alpha: T \rightarrow \mathbb{G}_{m,S}$ is faithfully flat and H is a separated presheaf, $-\alpha(t)^{-1}$ may be replaced by an arbitrary section of $\mathbb{G}_{m,S}$. Thus:

³⁷The editors corrected the calculation that follows.

Lemma 6.2.1. *For every $S' \rightarrow S$ and every $z \in \mathbb{G}_m(S')$,*

$$f_{-\alpha}(z\tilde{u}) = f_{\alpha}(z^{-1}u) f_T(\alpha^*(-z^{-1})) h f_{\alpha}(z^{-1}u).$$

Let $x, y \in \mathbb{G}_a(S')$, where $S' \rightarrow S$, and suppose that y and $1 + xy$ are invertible. Applying Lemma 6.2.1 first with $z = y$, one obtains³⁸

$$f_{\alpha}(xu) f_{-\alpha}(y\tilde{u}) = f_{\alpha}((x + y^{-1})u) f_T(\alpha^*(-y^{-1})) h f_{\alpha}(y^{-1}u).$$

Now $x + y^{-1} = y^{-1}(1 + xy)$. Applying the lemma with $z = y/(1 + xy)$ gives

$$\begin{aligned} f_{\alpha}((x + y^{-1})u) &= f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right) f_{\alpha}(-(x + y^{-1})u) \\ &\quad \cdot h f_T\left(\alpha^*\left(-\frac{y}{1 + xy}\right)\right). \end{aligned}$$

Substitution in the preceding identity yields

$$\begin{aligned} f_{\alpha}(xu) f_{-\alpha}(y\tilde{u}) &= f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right) f_{\alpha}(-(x + y^{-1})u) h^{-1} \\ &\quad \cdot f_T(\alpha^*(1 + xy)^{-1}) h f_{\alpha}(y^{-1}u). \end{aligned}$$

Since $h^{-1}f_T(t)h = f_T(s_{\alpha}(t))$ by (2) (recall that $s_{\alpha}^2 = \text{id}$), and $s_{\alpha} \circ \alpha^* = -\alpha^*$, this is

$$f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right) f_{\alpha}(-(x + y^{-1})u) f_T(\alpha^*(1 + xy)) f_{\alpha}(y^{-1}u).$$

For $x' \in U_{\alpha}(S')$ and $z \in \mathbb{G}_m(S')$, one has

$$f_{\alpha}(x') f_T(\alpha^*(z)) = f_T(\alpha^*(z)) f_{\alpha}(z^{-2}x').$$

Consequently,

$$\begin{aligned} f_{\alpha}(xu) f_{-\alpha}(y\tilde{u}) &= f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right) f_T(\alpha^*(1 + xy)) \\ &\quad \cdot f_{\alpha}\left(\left[-\frac{y^{-1}(1 + xy)}{(1 + xy)^2} + y^{-1}\right]u\right) \\ &= f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right) f_T(\alpha^*(1 + xy)) f_{\alpha}\left(\frac{x}{1 + xy}u\right). \end{aligned}$$

We have proved:

Lemma 6.2.2. *Let $S' \rightarrow S$. If $a \in U_{\alpha}(S')$, $b \in U_{-\alpha}^{\times}(S')$, and $1 + ab \in \mathbb{G}_m(S')$, then*

$$f_{\alpha}(a) f_{-\alpha}(b) = f_{-\alpha}\left(\frac{b}{1 + ab}\right) f_T(\alpha^*(1 + ab)) f_{\alpha}\left(\frac{a}{1 + ab}\right).$$

By schematic density, this formula remains valid for $b \in U_{-\alpha}(S')$, with $1 + ab$ invertible. Define

$$f: \Omega \longrightarrow H, \quad f(ytx) = f_{-\alpha}(y) f_T(t) f_{\alpha}(x).$$

³⁸The editors expanded the calculations that follow.

It follows at once from Lemma 6.2.2, condition 6.2(i), and formula (F') of 2.4 that whenever $g, g' \in \Omega(S')$ and $gg' \in \Omega(S')$, one has

$$f(gg') = f(g)f(g').$$

By Exposé XVIII, 2.3(iii) and 2.4,³⁹ there is therefore a group morphism $G \rightarrow H$ extending f . Denote it again by f . It remains only to prove that $f(w_\alpha) = h$. Since

$$w_\alpha = u \cdot (-\tilde{u}) \cdot u,$$

formula (*₃) gives⁴⁰

$$f(w_\alpha) = f_\alpha(u)f_{-\alpha}(-\tilde{u})f_\alpha(u) = h.$$

This completes the proof of Theorem 6.2.

Remark 6.3. *These results will be completed in Exposé XXIII, 3.5.*

³⁹Each geometric fiber of G is connected, for example by 1.1.

⁴⁰The editors simplified the original argument by invoking (*₃).